上海科技大学
ShanghaiTech University

SI100B Introduction to Information Science and Technology

Electrical Engineering

Instructor: Dr. Xin Lou

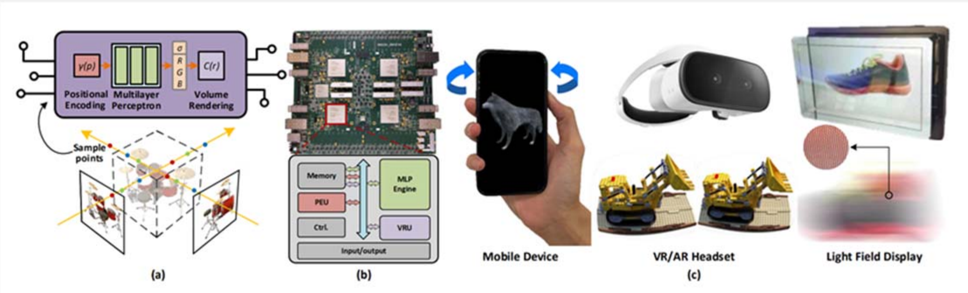
Email: louxin@shanghaitech.edu.cn

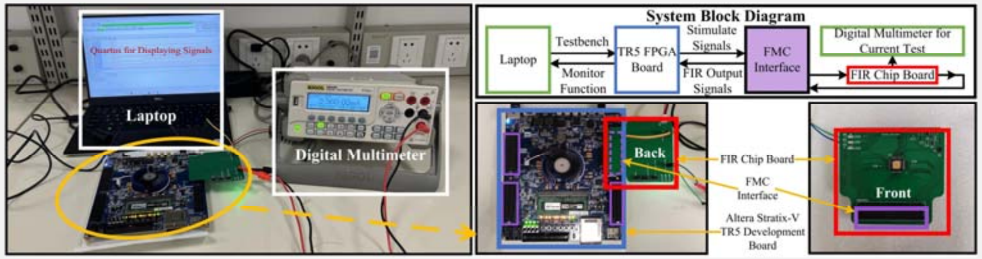
Office: Room 314, SIST Building #3

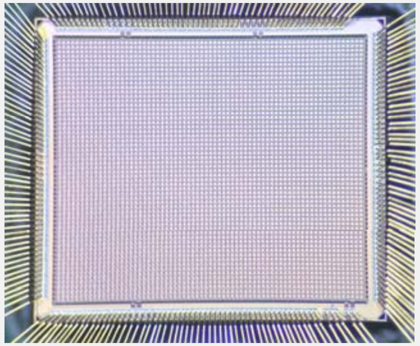
VLSI Signal Processing Lab

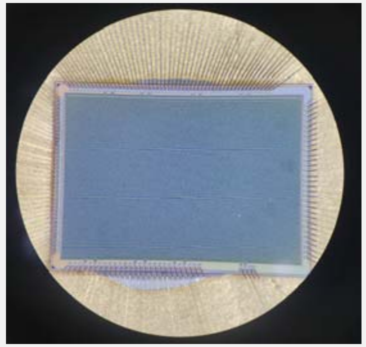
上海科技大学 ShanghaiTech University

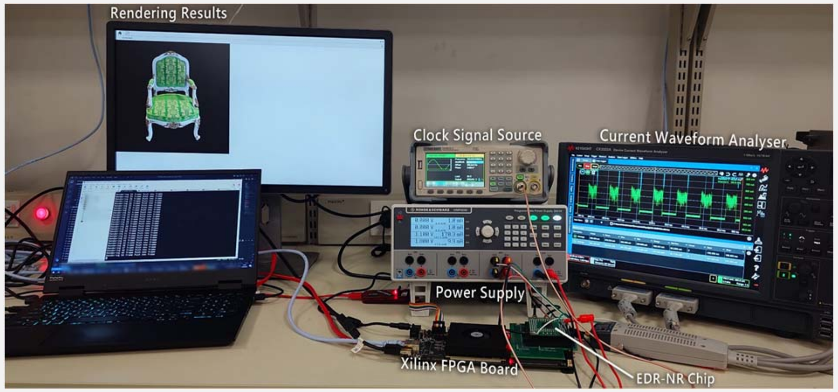
- We conduct research on **integrated circuits and systems** for various **image, graphics and vision** processing applications
- We build prototypes based on custom IC chip design, FPGA, as well as other embedded systems





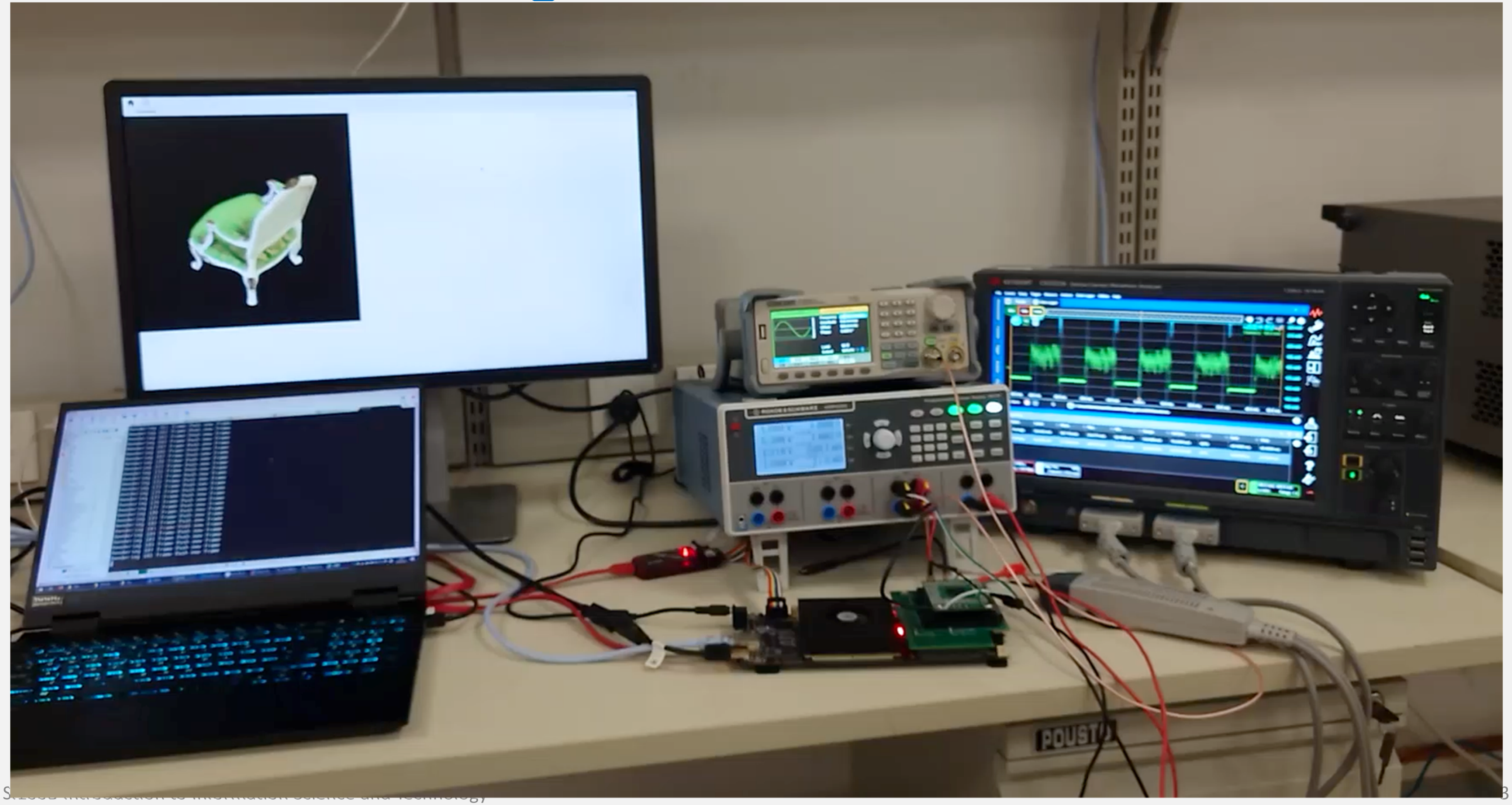






Neural Rendering Demo

上海科技大学
ShanghaiTech University



Assessment

➢ Two Lab Assignments: 10% for each

Reference Books

- Sarah Harris and David Harris, Digital Design and Computer Architecture, RISC-V Edition: RISC-V Edition, Morgan Kaufmann, 2021

Lecture 1: Introduction

The Three Pillars of AI

General Architecture of AI Systems

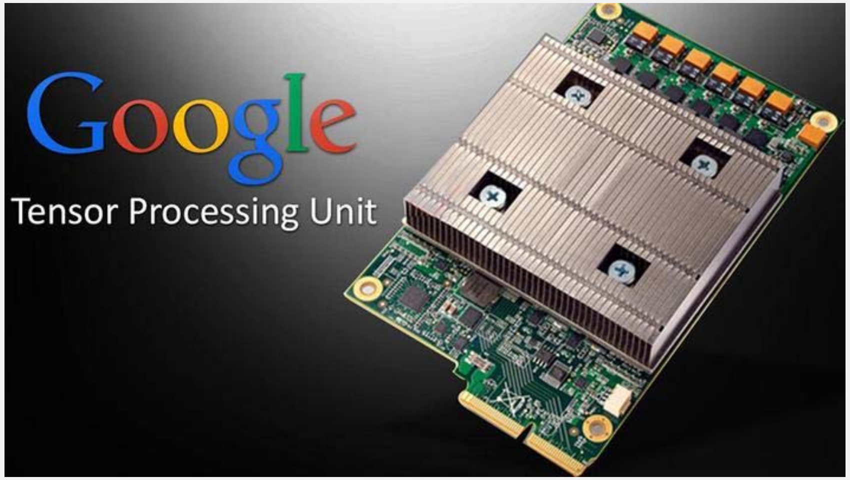
图片来源：《AI系统：原理与架构》

Computation Devices for AI

CPU

GPU

FPGA



ASIC

Application of AI Chips

上海科技大学
ShanghaiTech University

云端训练

- **可部署芯片**: GPU/GPU/ASIC
- **芯片特征**：高吞吐量、高精确率、可编程性、分布式、可扩展性、高内存与带宽
- **计算能力与功耗**: >30TOPS，>50W
- **应用**：云/HPC/数据中心

云端推理

- **可部署芯片**: GPU/GPU/ASIC/FPGA
- **芯片特征**：高吞吐量、高精确率、分布式、可扩展性、低延时
- **计算能力与功耗**: 30TOPS，>50W
- **应用**：云/HPC/数据中心

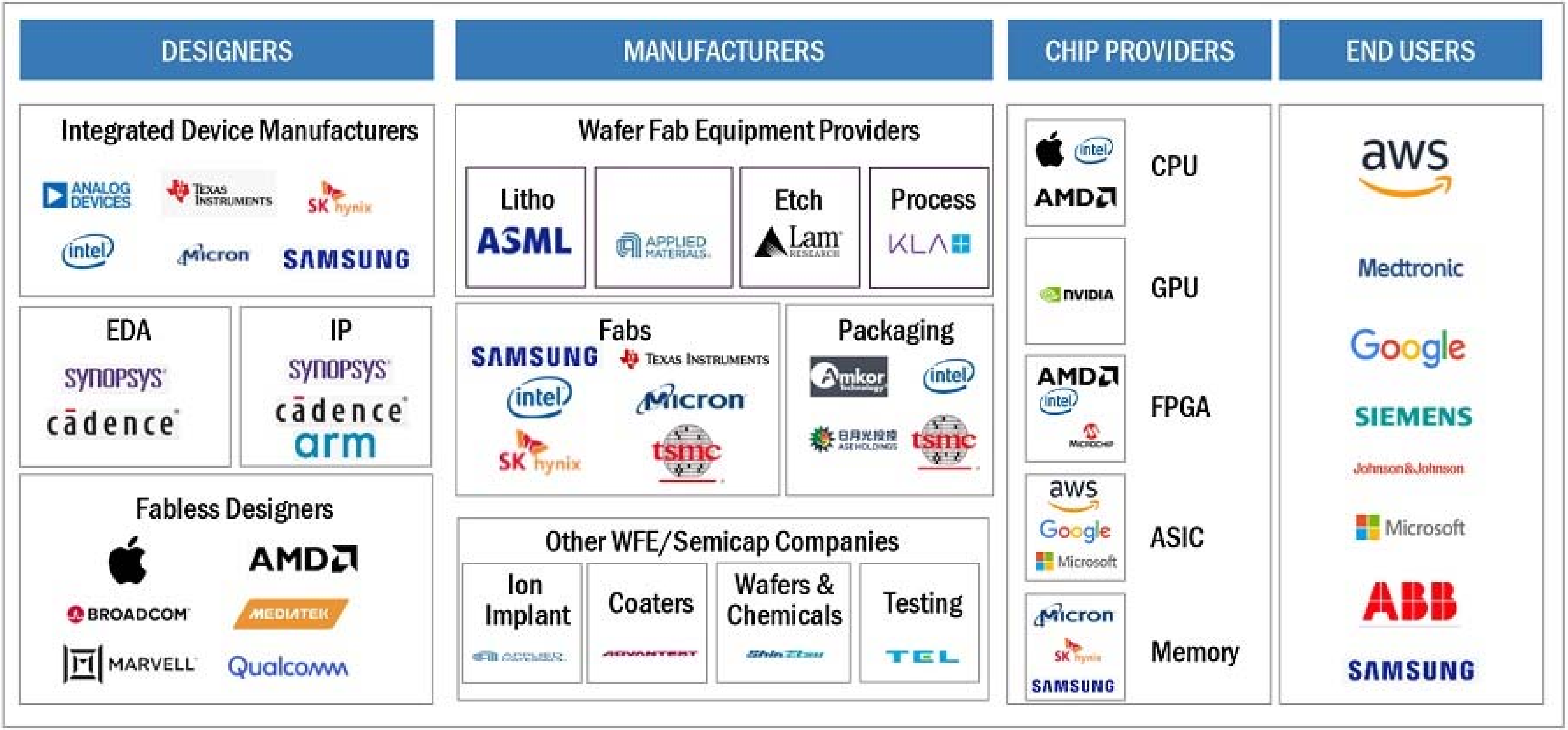
边缘计算

- **可部署芯片**: GPU/GPU/ASIC/FPGA
- **芯片特征**：降低AI计算延迟、可单独部署或与其他设备组合（如5G基站）、可将多个终端用户进行虚拟化、较小的机架空间、扩展性及加速算法
- **计算能力与功耗**: 5~30TOPS，4~15W
- **应用**：智能制造、智慧家居、智慧交通等、智慧金融等众多领域

终端设备

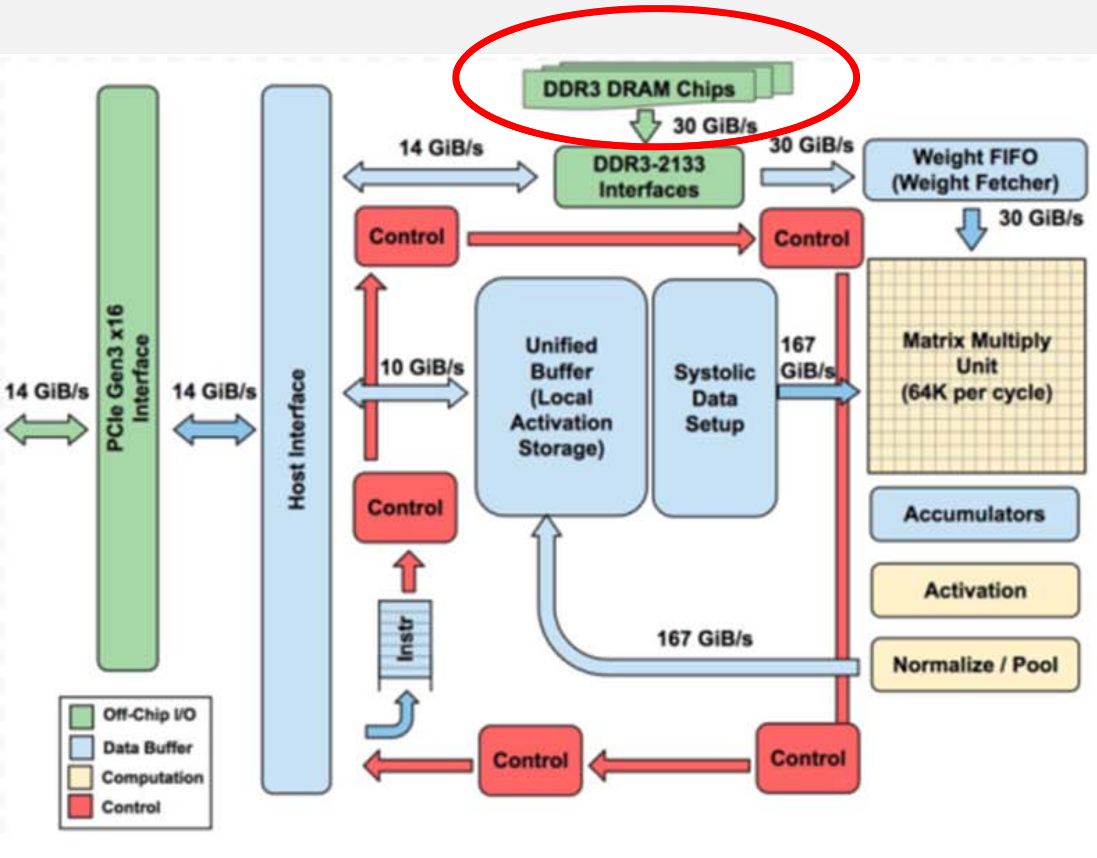
- **可部署芯片**: GPU/GPU/ASIC/FPGA
- **芯片特征**：低功耗、高能效、推理任务为主、较低的吞吐量、低延迟、成本敏感
- **计算能力与功耗**: <8TOPS，<5W
- **应用**：各类消费电子，产品形态多样；以及物联网领域

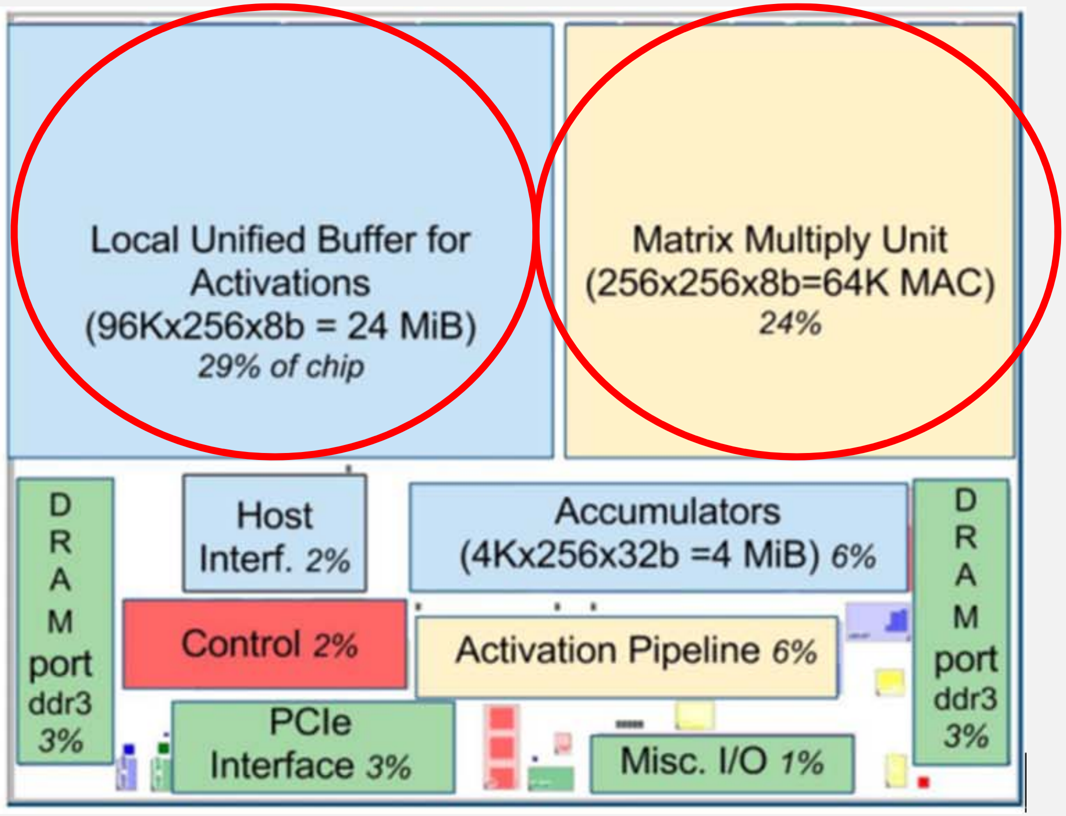
AI Chip Landscape



Look into the AI Chips-Memory & Logic

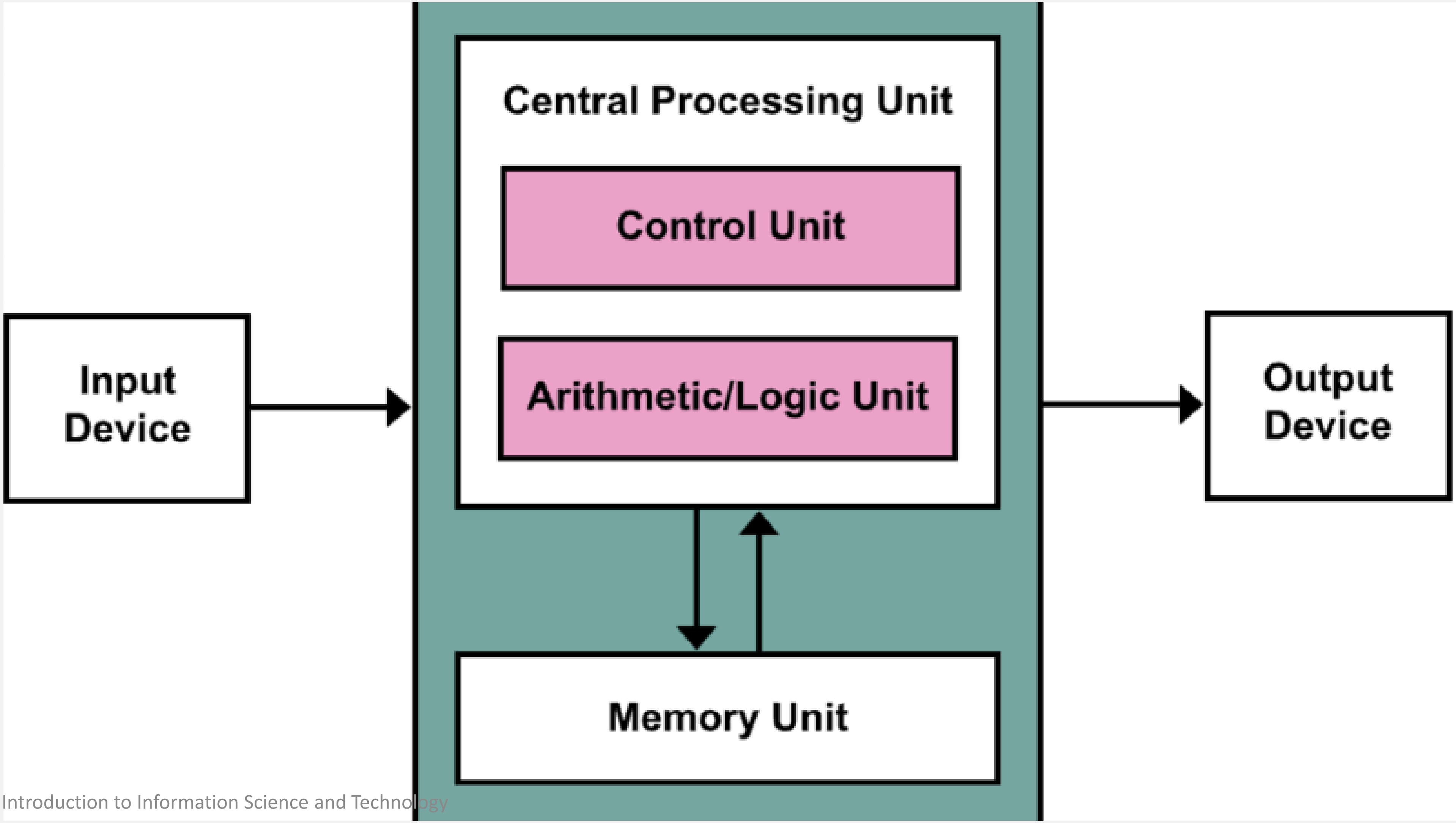
上海科技大学 ShanghaiTech University





The TPU Architecture & Floorplan of the TPU die.

Components of a Computer



Python / Application

High Level Language Program (e.g., C)

Compiler

Assembly Language Program (e.g., RISC-V)

Assembler

Machine Language Program (RISC-V)

Machine Interpretation

Hardware Architecture Description (e.g., block diagrams)

Architecture Implementation

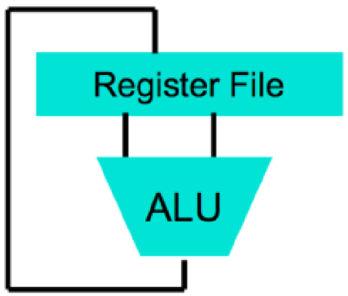
Logic Circuit Description (Circuit Schematic Diagrams)

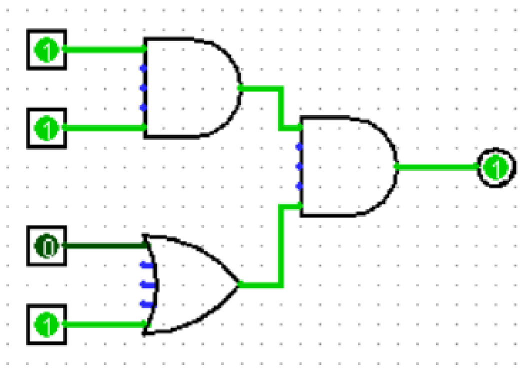
Physics

```
temp = v[k];
v[k] = v[k+1];
v[k+1] = temp;
```

```
lw   t0, 0(s2)
lw   t1, 4(s2)
sw   t1, 0(s2)
sw   t0, 4(s2)
```

```
0000 1001 1100 0110 1010 1111 0101 1000
1010 1111 0101 1000 0000 1001 1100 0110
1100 0110 1010 1111 0101 1000 0000 1001
0101 1000 0000 1001 1100 0110 1010 1111
```





For details, check following course:

CS110 Computer Architecture I

Anything can be represented as a *number*, i.e., data or instructions

Memory

A component where instructions and data are stored.

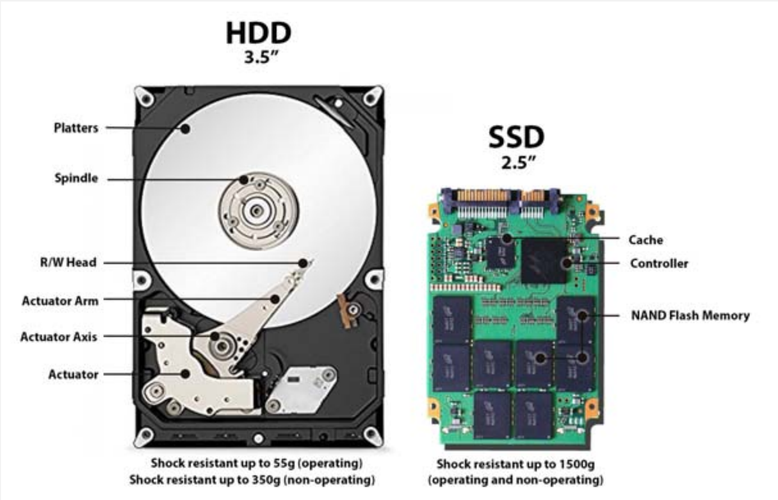
Memory

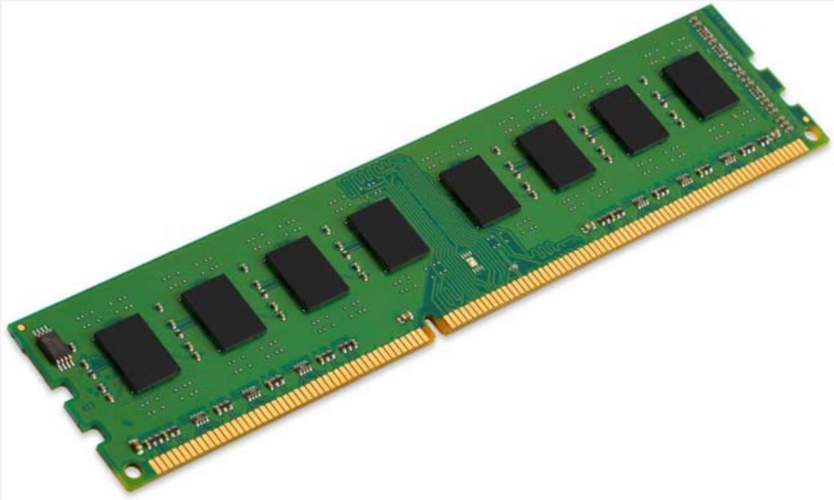
SK Hynix's Wild Rise Triggers Exchange Caution
SK hynix Inc - Last Price
+219.59% (+220.61%ann.)
0.6M
0.5M
0.4M
0.3M
0.2M
Korean Won
Nov
Dec
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
2024
2025
Source: Bloomberg
Bloomberg

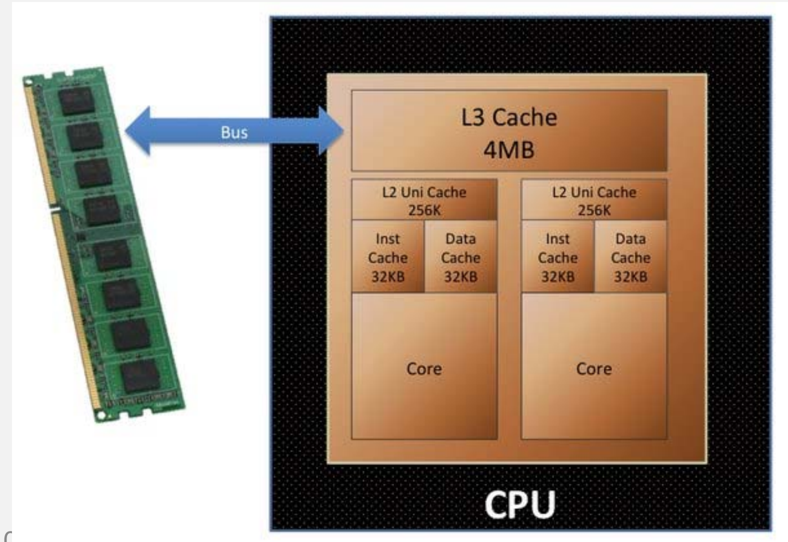
Memory

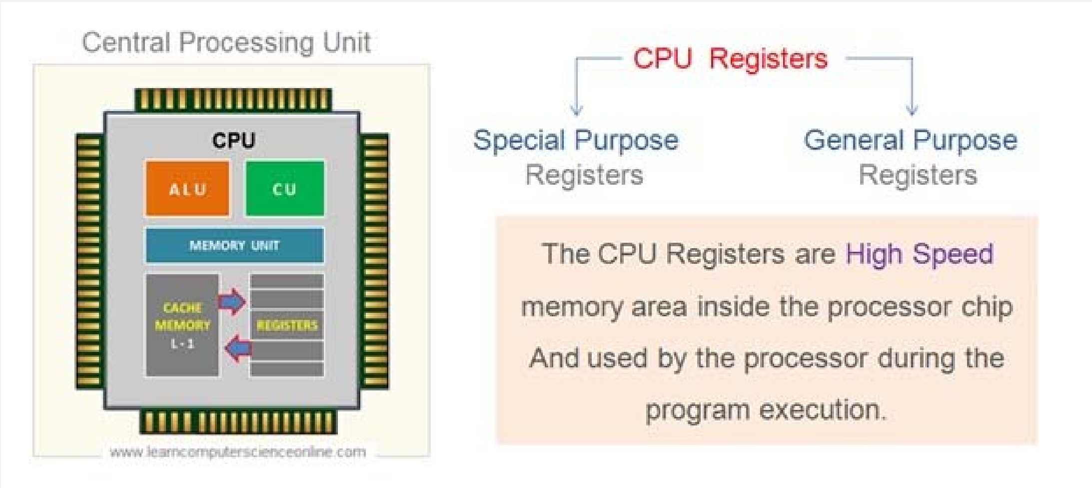
237.500
220.000
200.000
180.000
160.000
140.000
120.000
100.000
80.000
239.880
73.385
E
D
成交量
VOL: 2861.739
5000.00
萬
2025/05
06
07
08
09
10
11

Memory Types



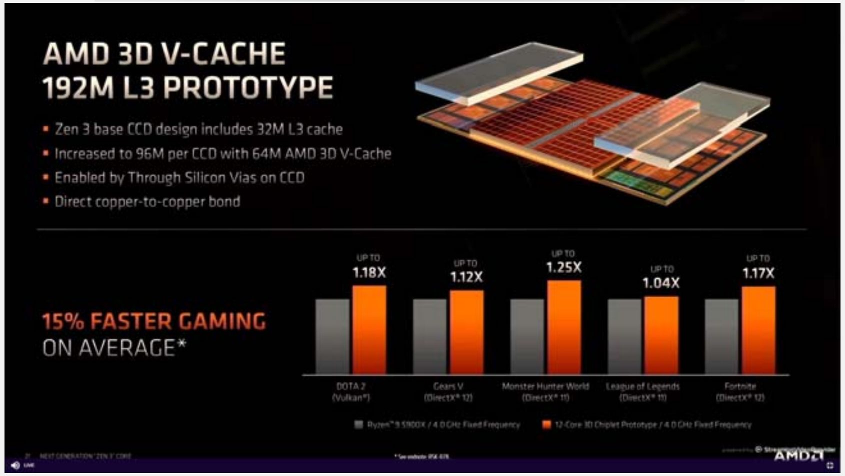




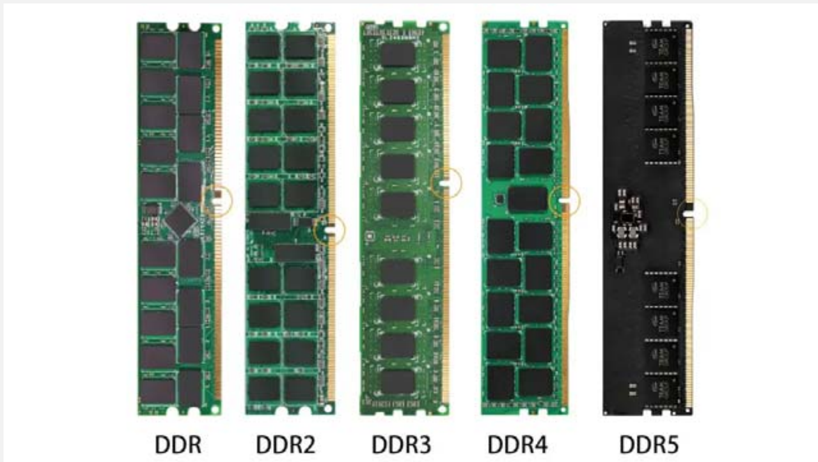


Capacity

Speed (Bandwidth)



All	Read (MB/s)	Write (MB/s)
SEQ1M Q8T1	6963.45	6533.18
SEQ128K Q32T1	6956.90	6541.99
RND4K Q32T16	3885.00	2993.73
RND4K Q1T1	94.92	335.35

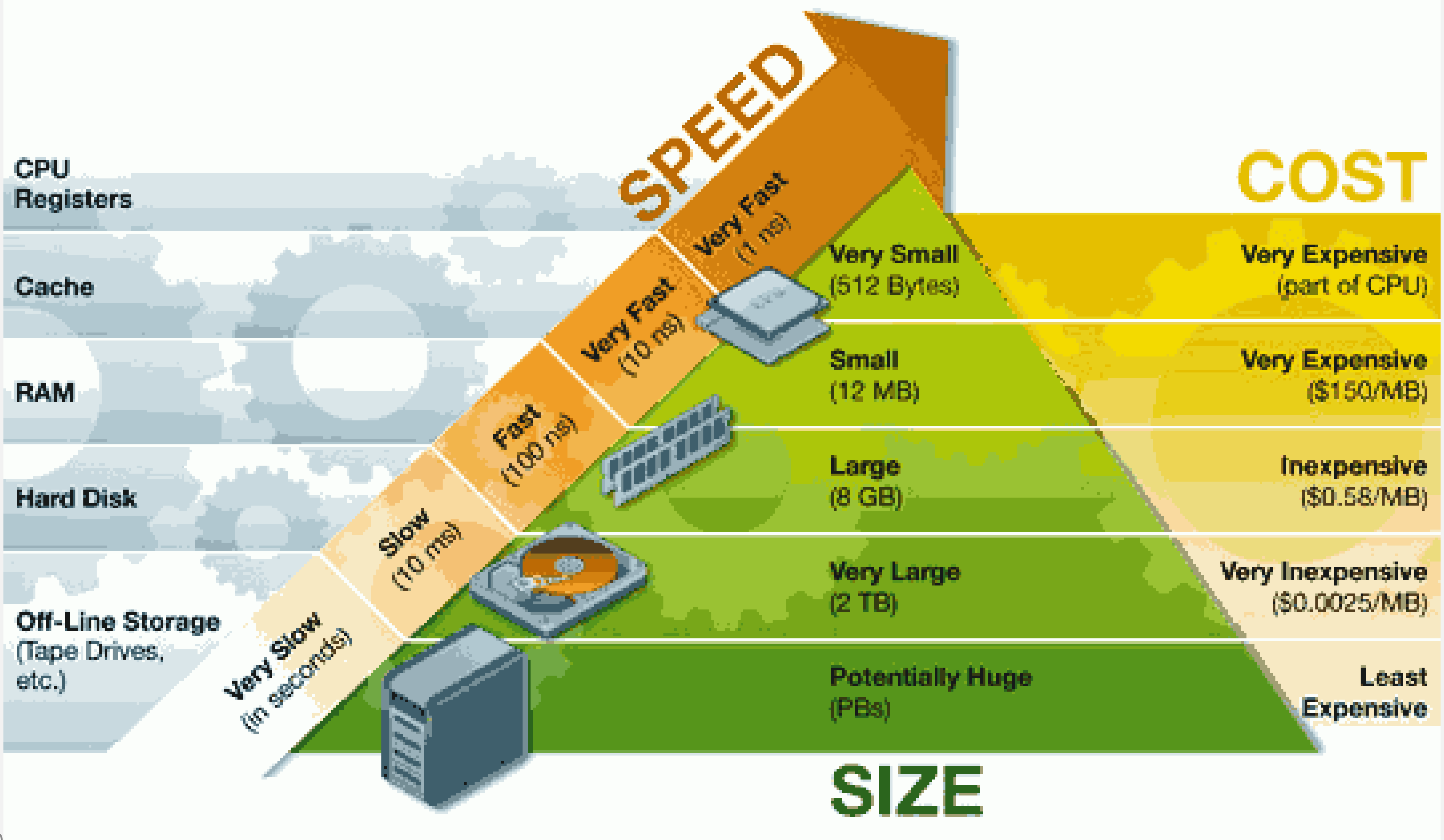


What do you want as a programmer?

Capacity

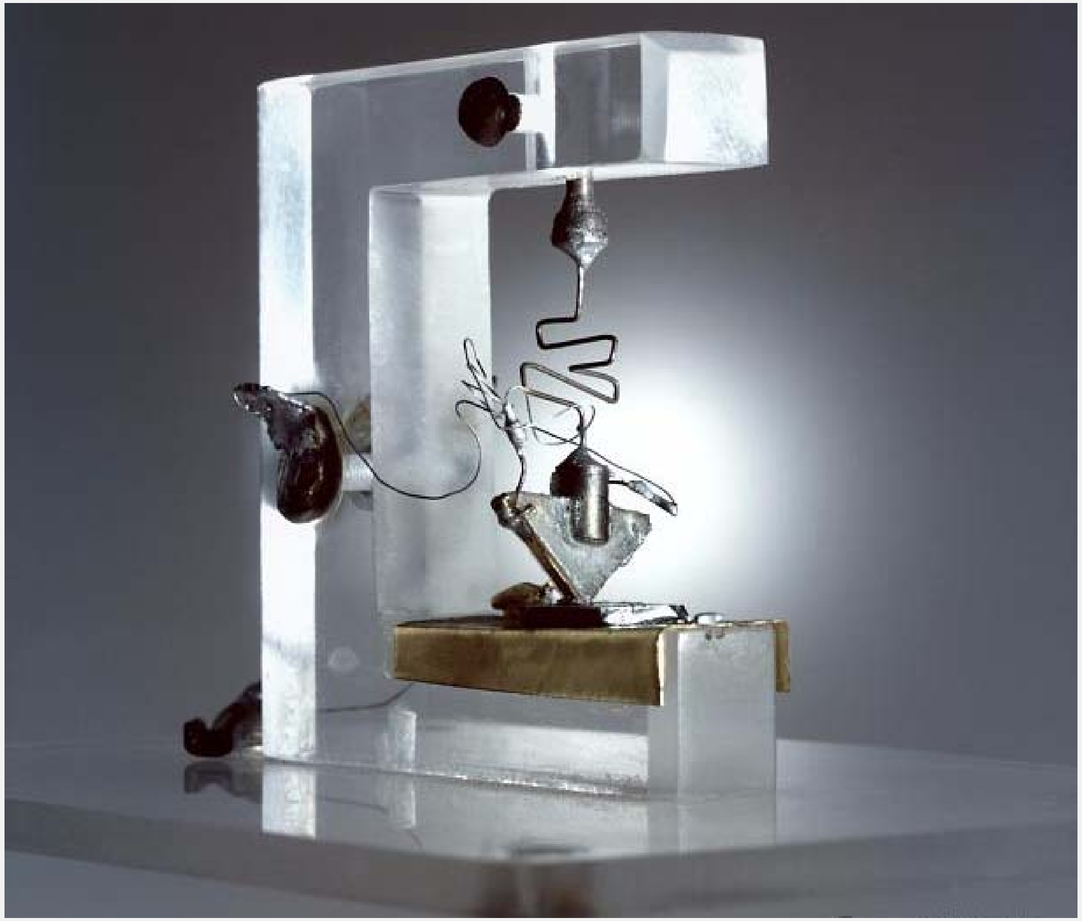
Speed (Bandwidth)

上海科技大学
ShanghaiTech University

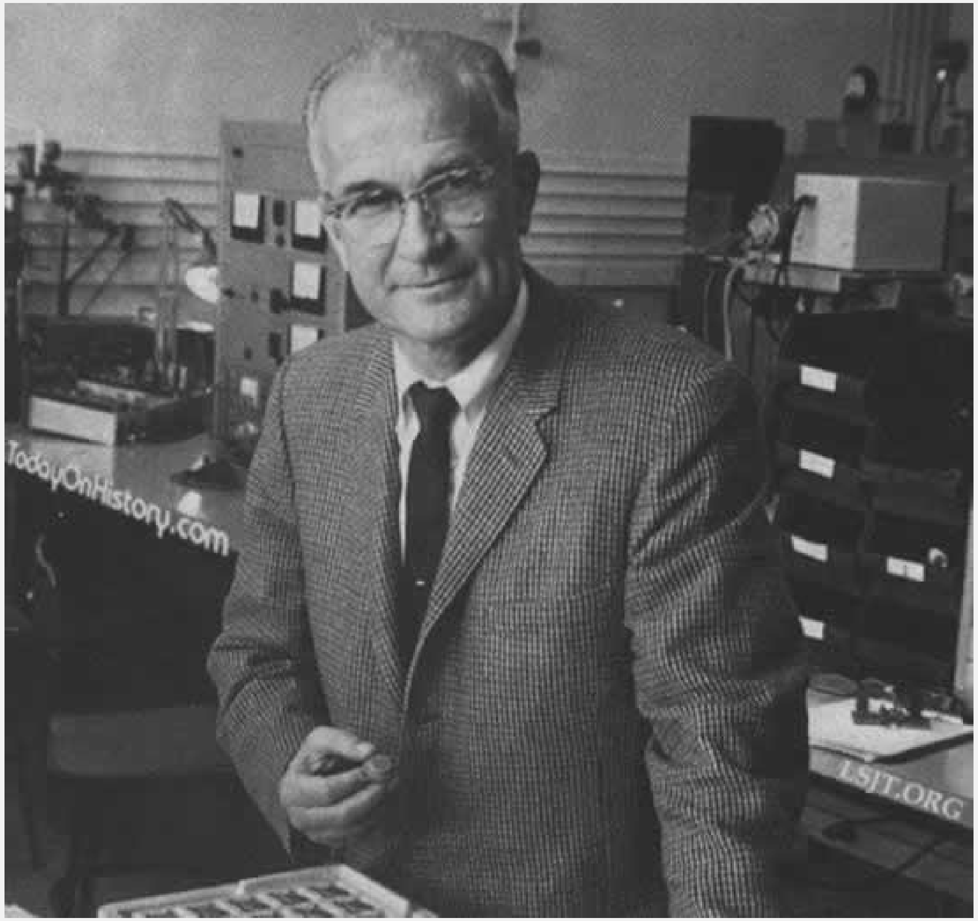


Transistor

The First Transistor

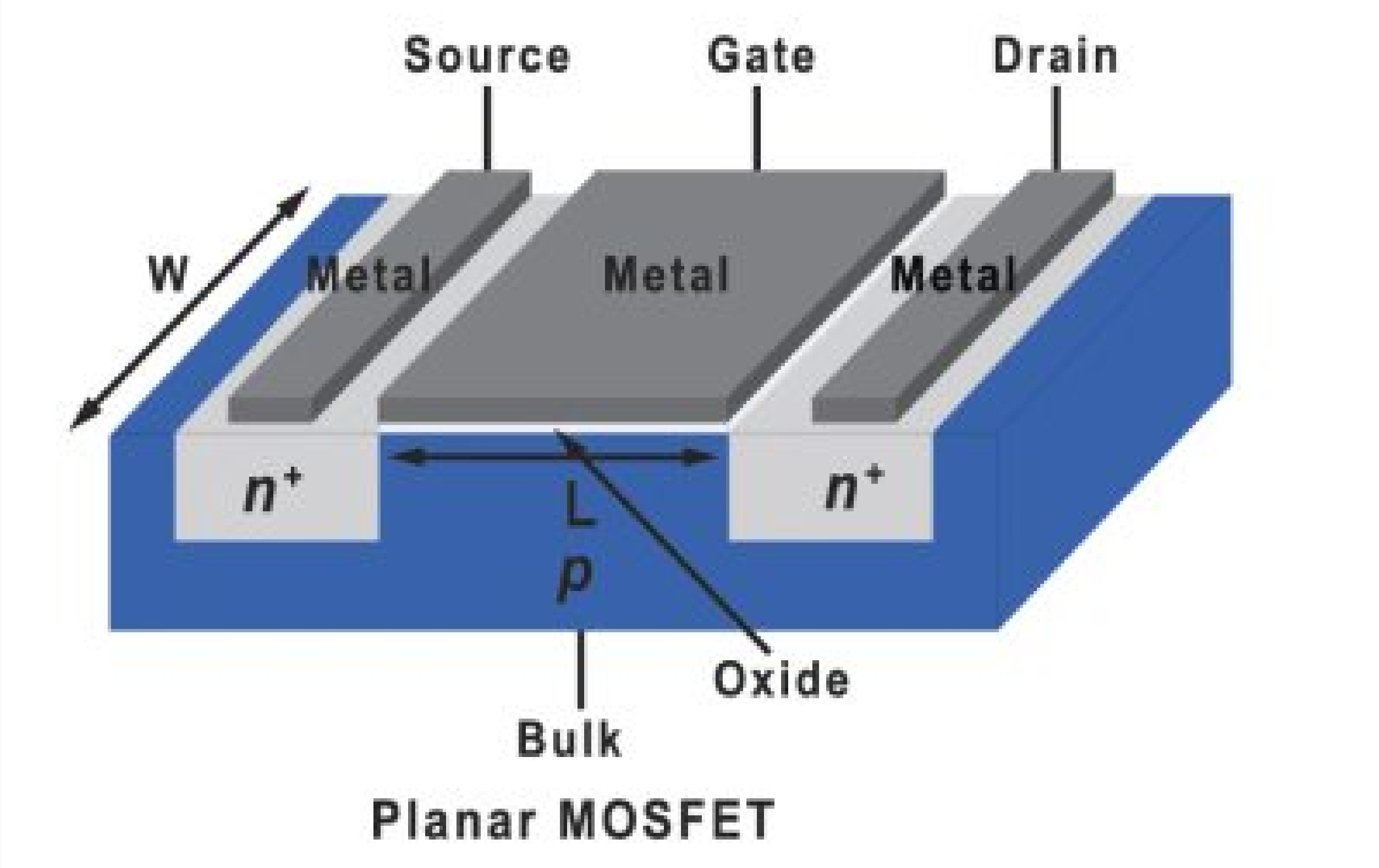


First Transistor, Bell Labs, 1947



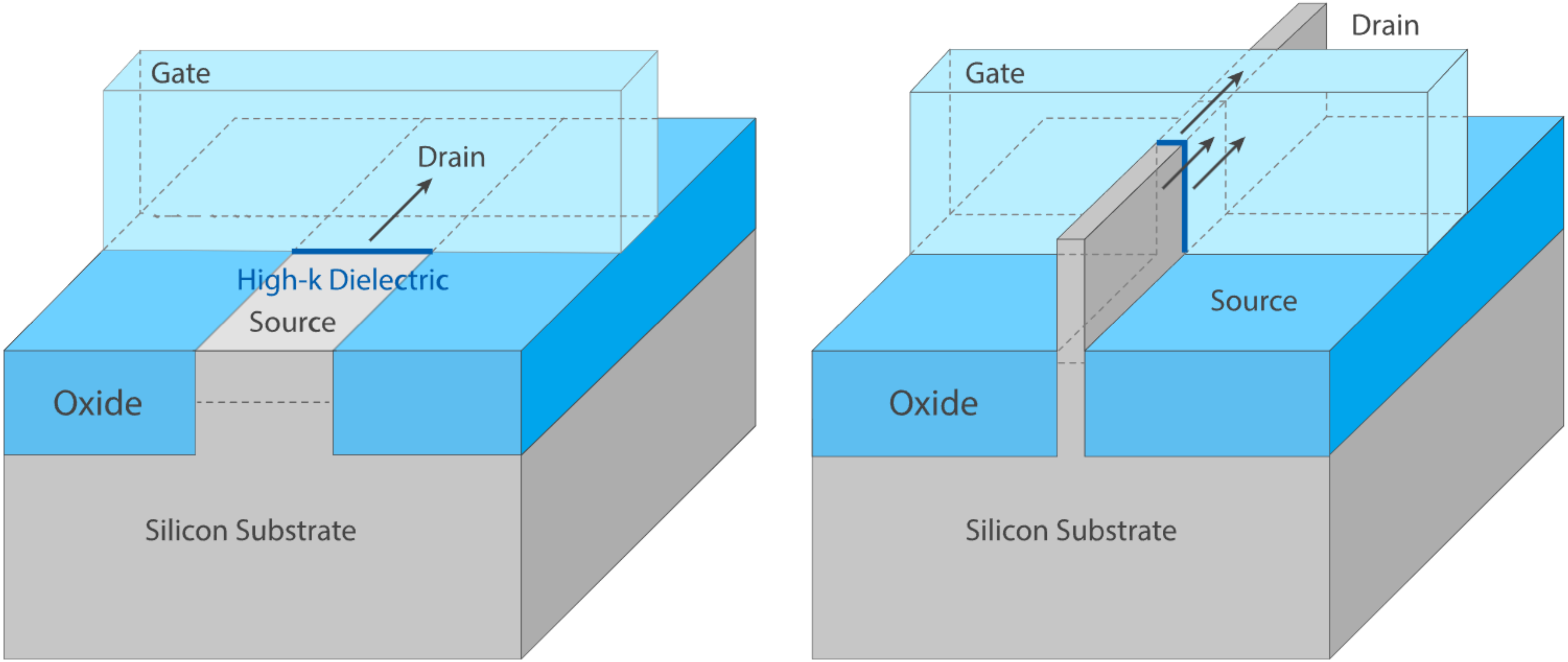
威廉·肖克利(William Shockley)

Nobel Prize for Physics in 1956


Source
Gate
Drain
W
Metal
Metal
Metal
n+
L
n+
p
Oxide
Bulk
Planar MOSFET

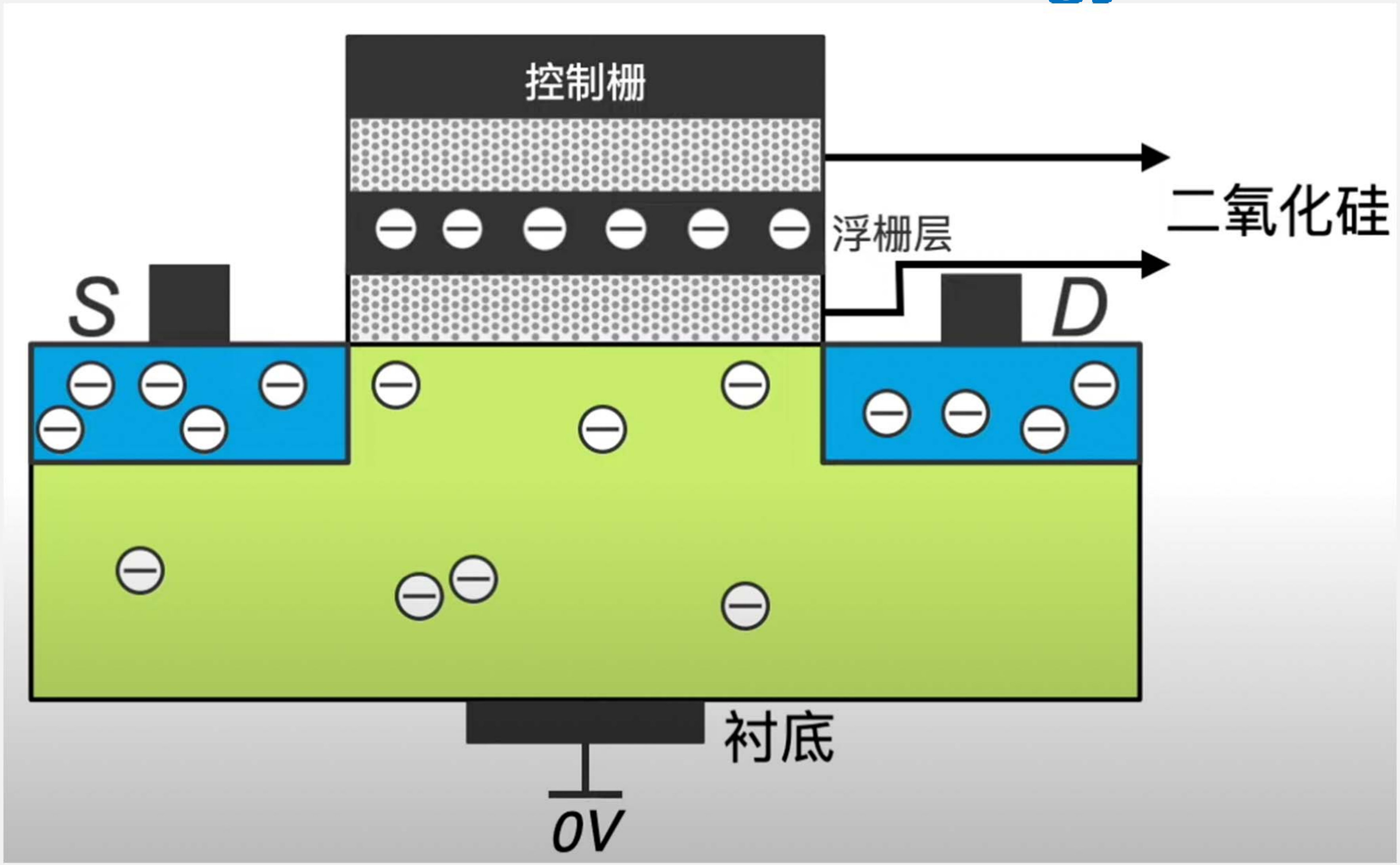
For details, check following course:

EE120 Fundamentals of Semiconductor Devices



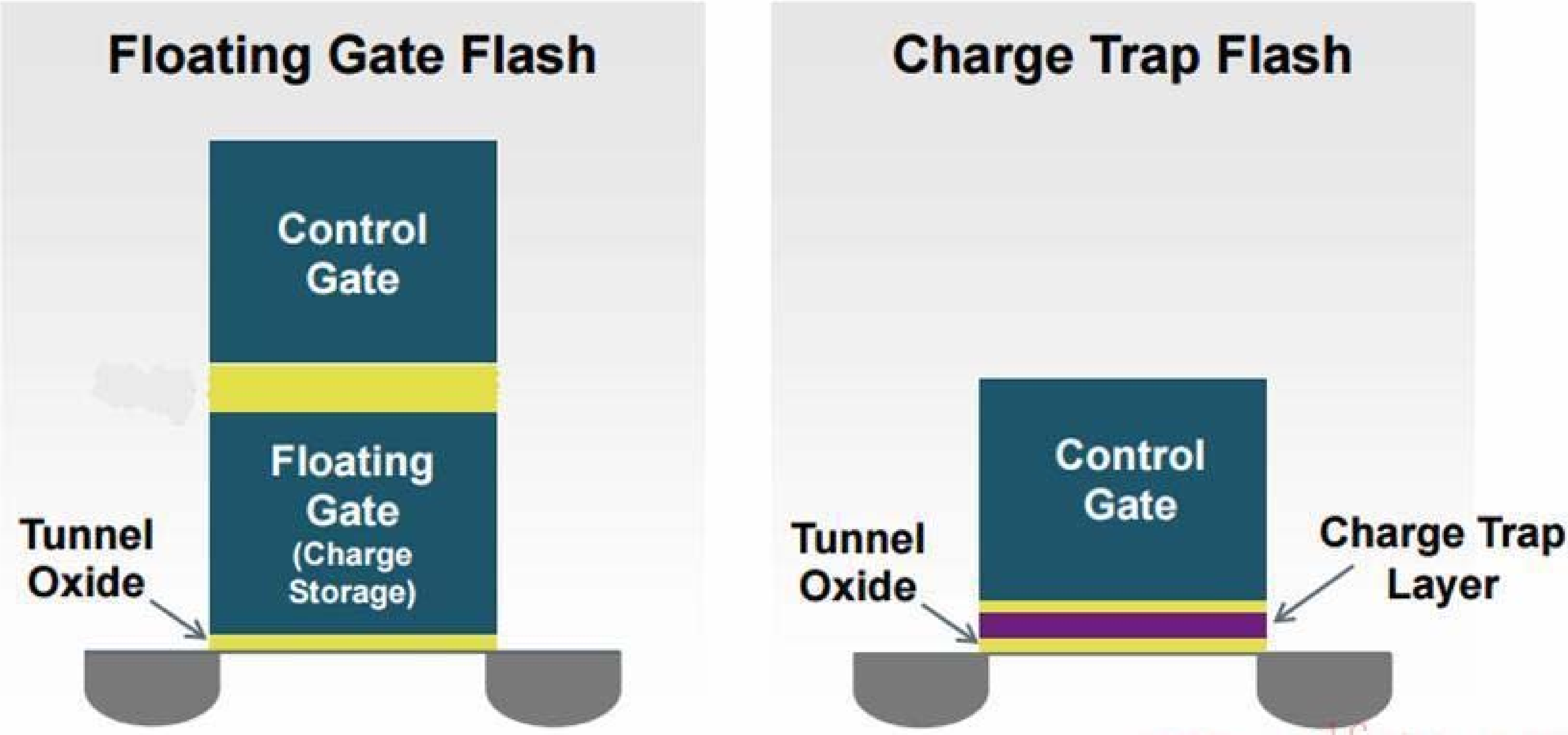
For details, check following course:
EE120 Fundamentals of Semiconductor Devices

Solid State Drive: Flash Technology

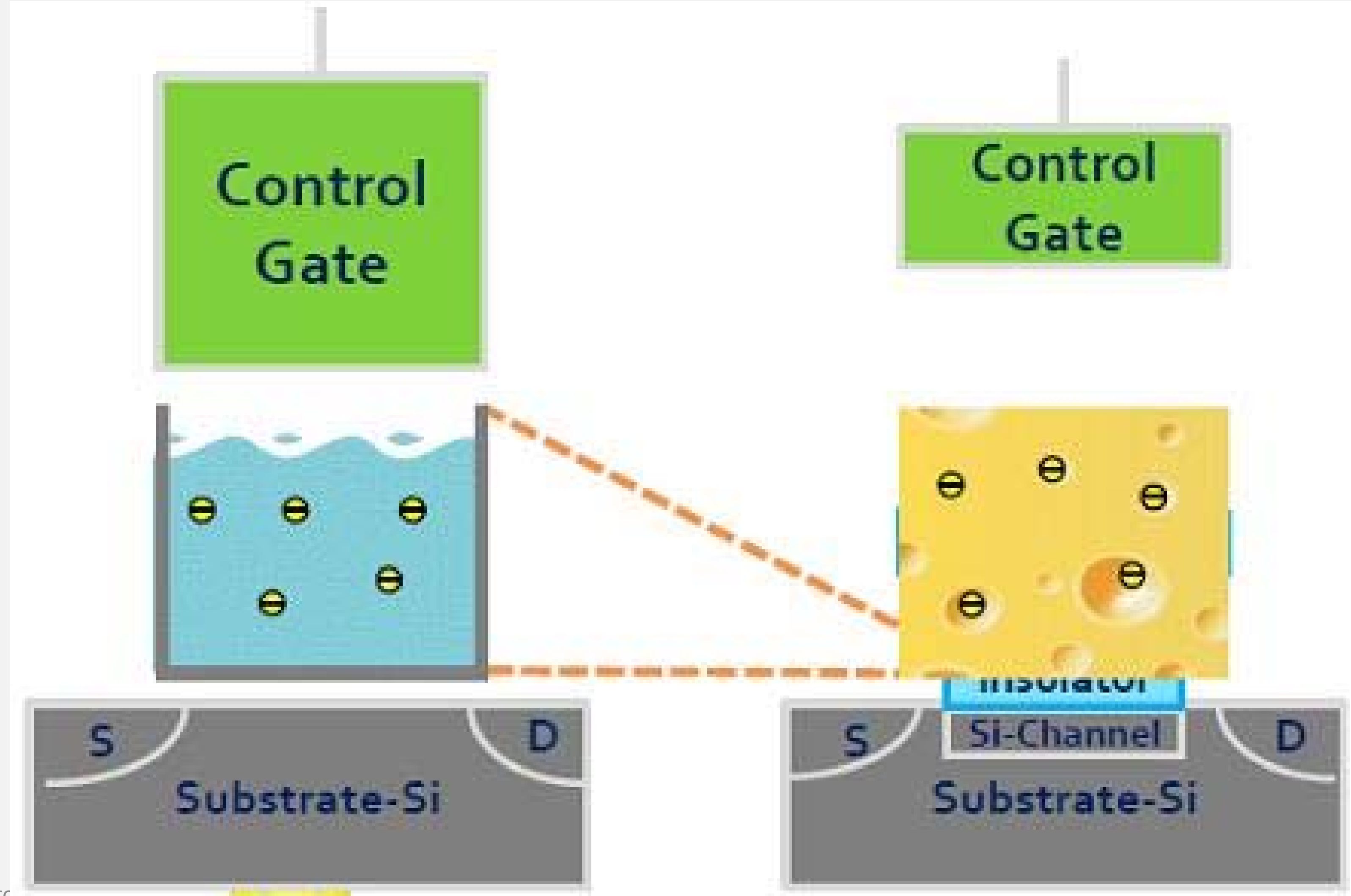


https://www.bilibili.com/video/BV1jw48eMED9/?spm_id_from=333.337.search-card.all.click&vd_source=9102d8c932f502e6a9f7e1a63a777959

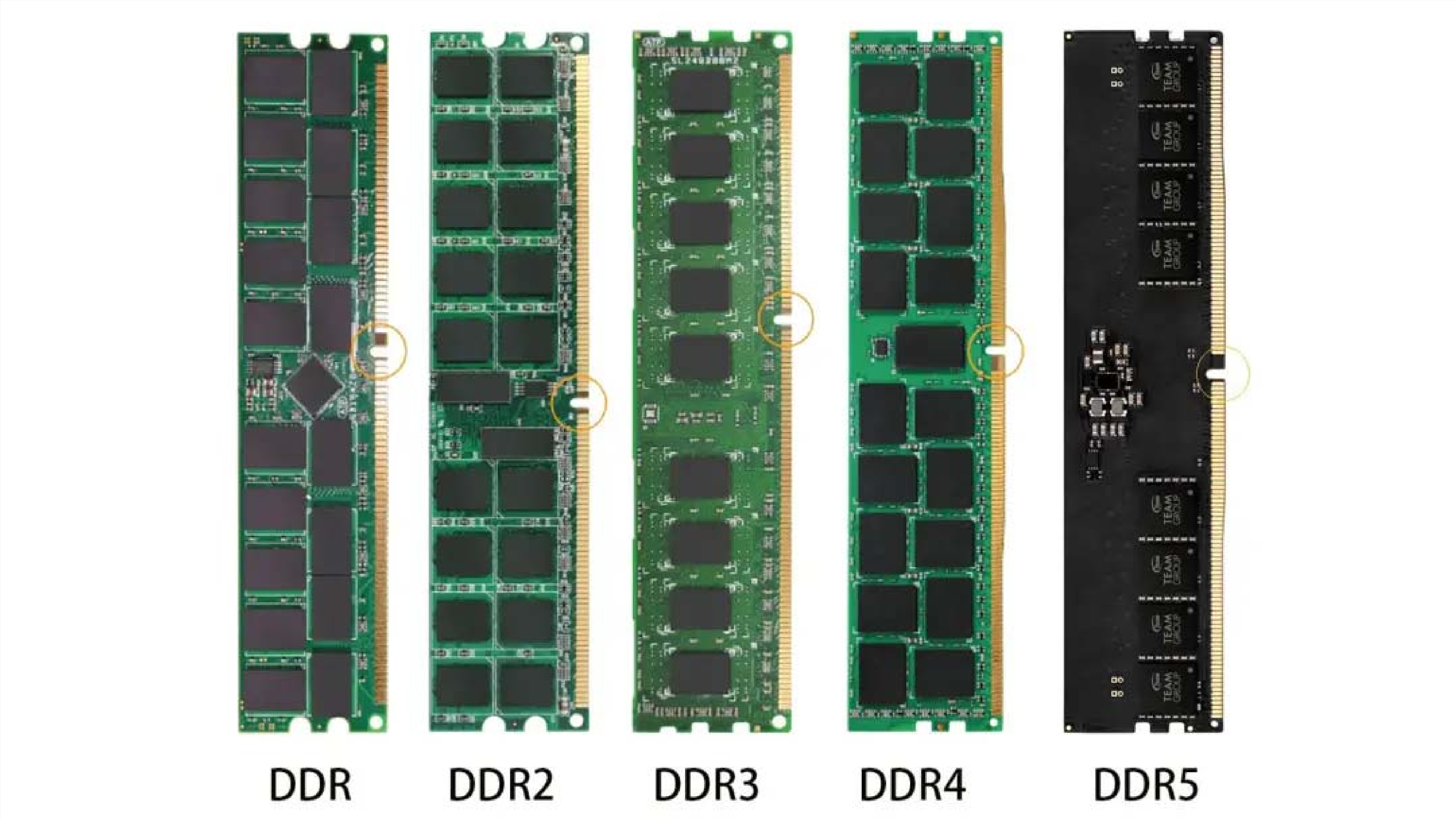
Solid State Drive: Flash Technology



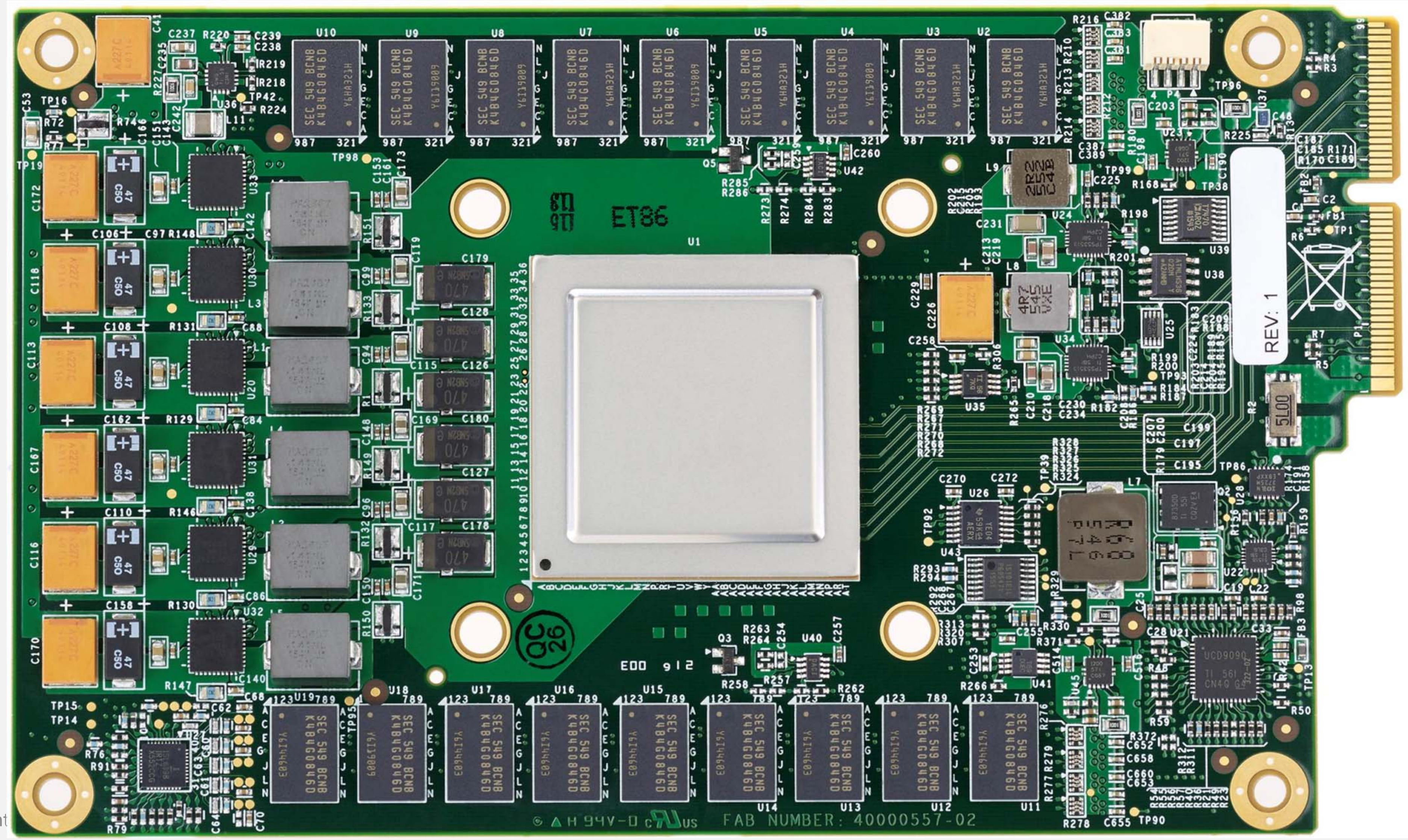
Solid State Drive: Flash Technology



Main Memory: DDR DRAM

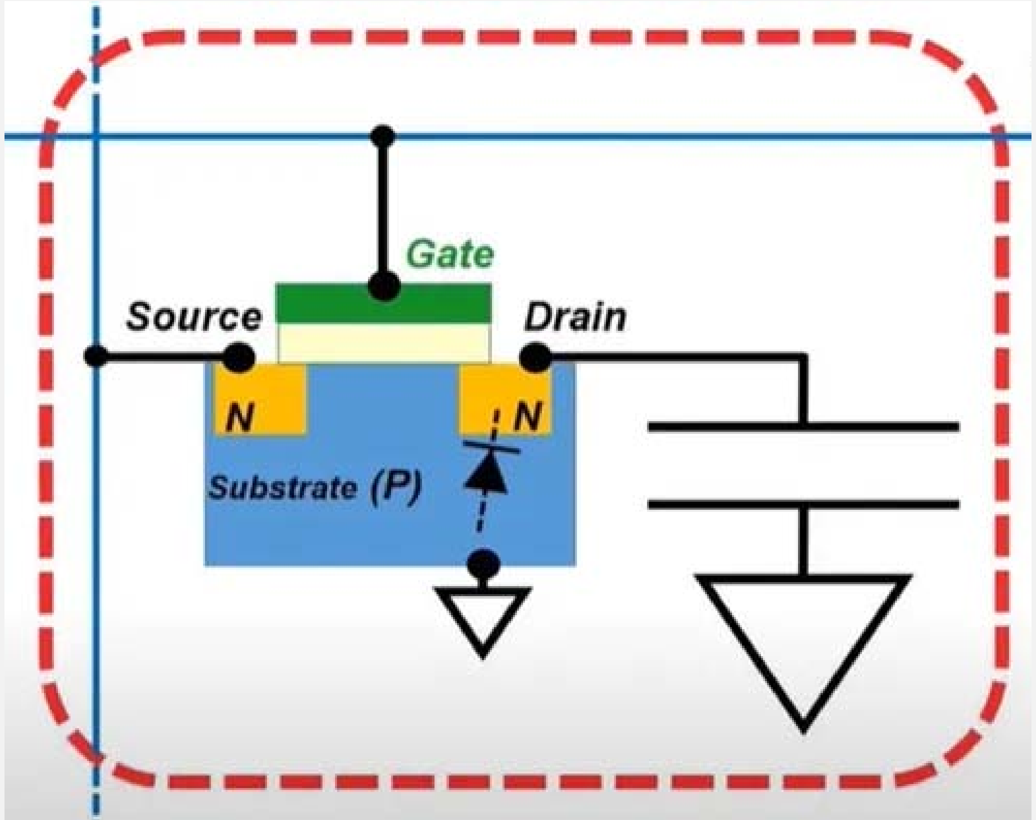


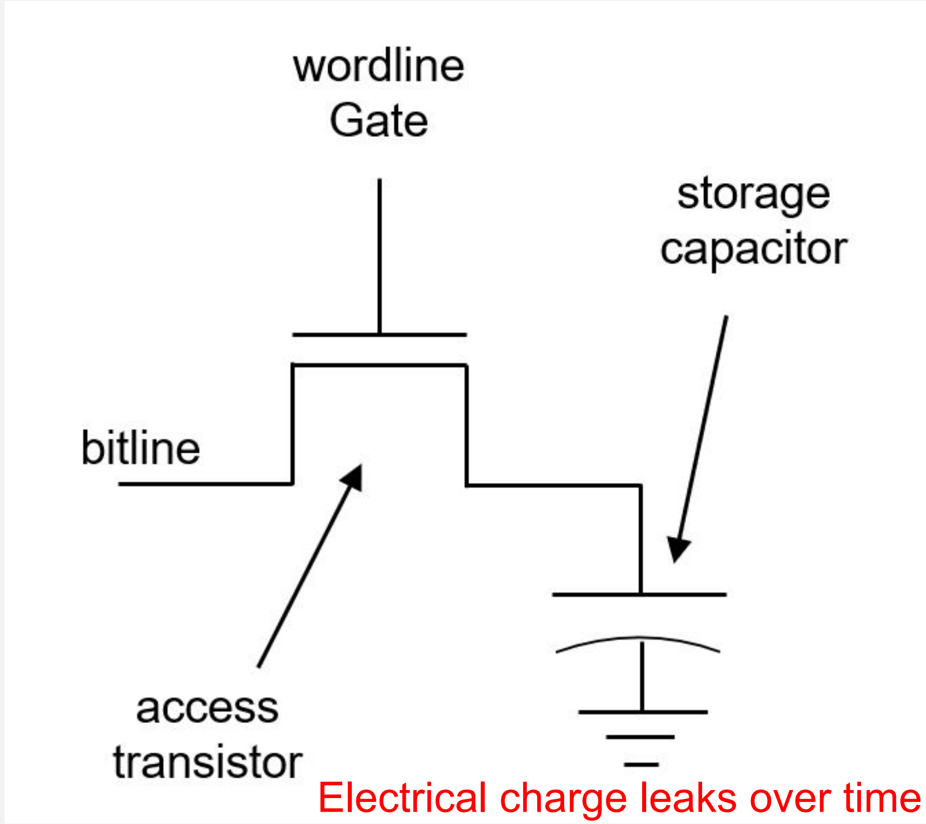
The TPU Print Circuit Board



DRAM Cell Structure

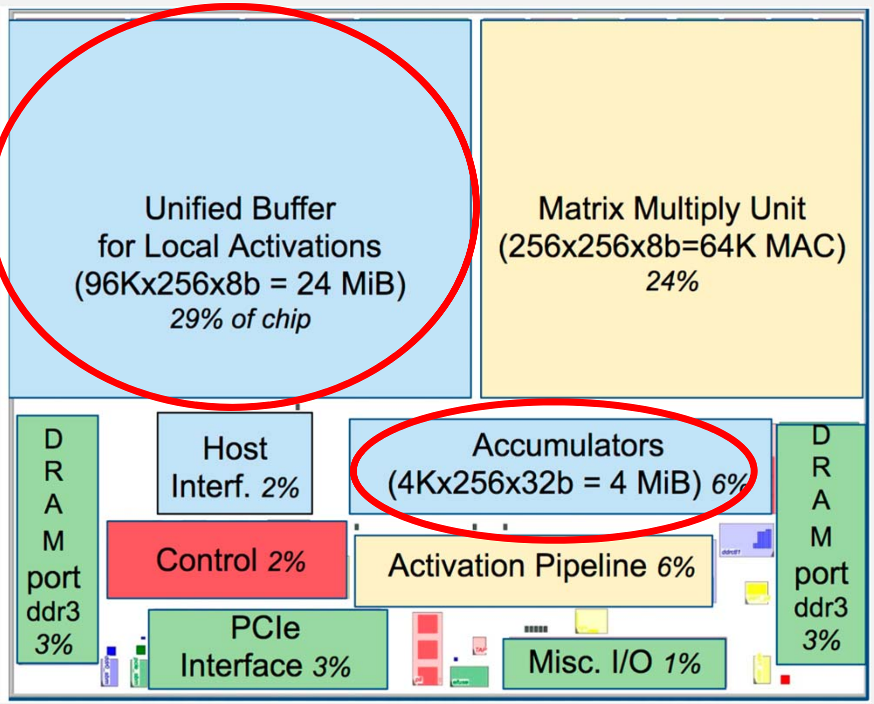
For details, check following course:
EE113 Digital Integrated Circuit Design

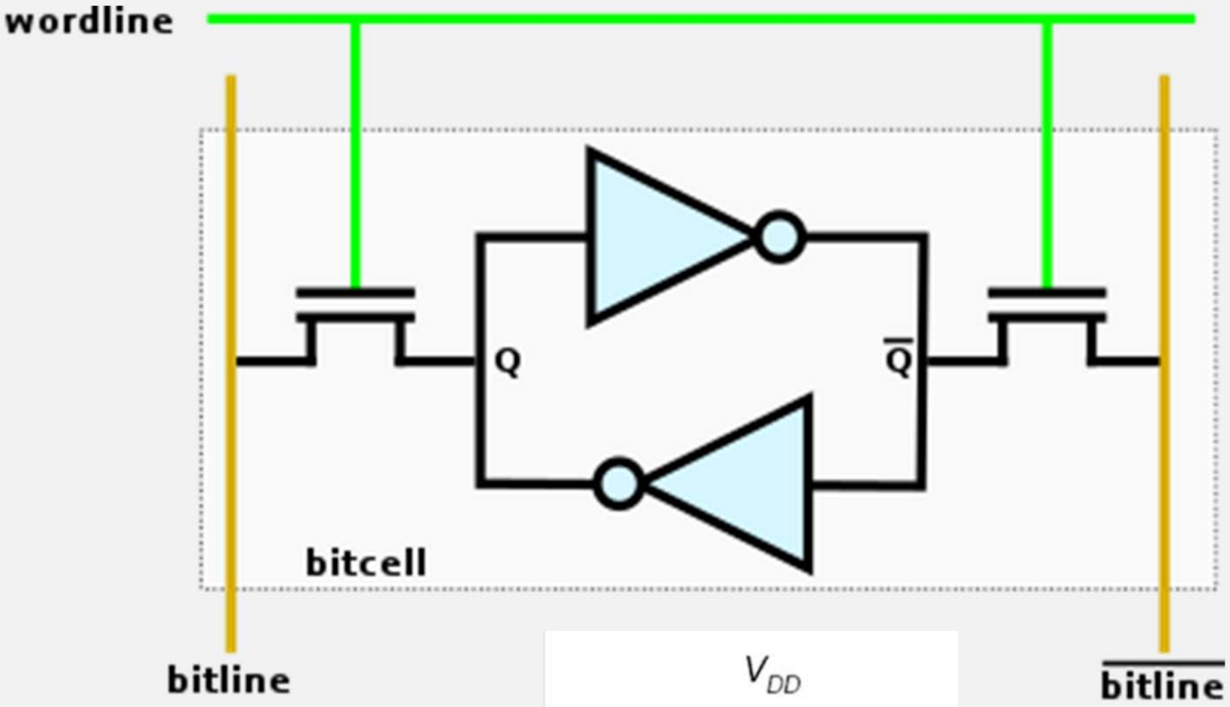


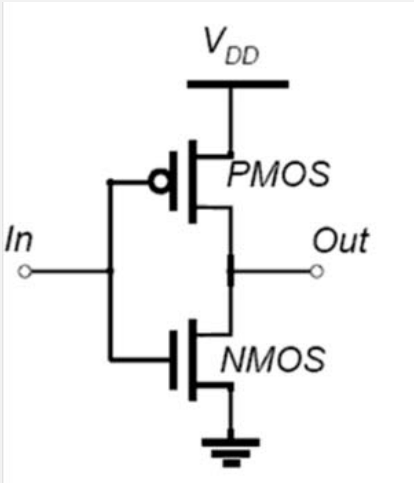


DRAM must be refreshed periodically to preserve the stored data!

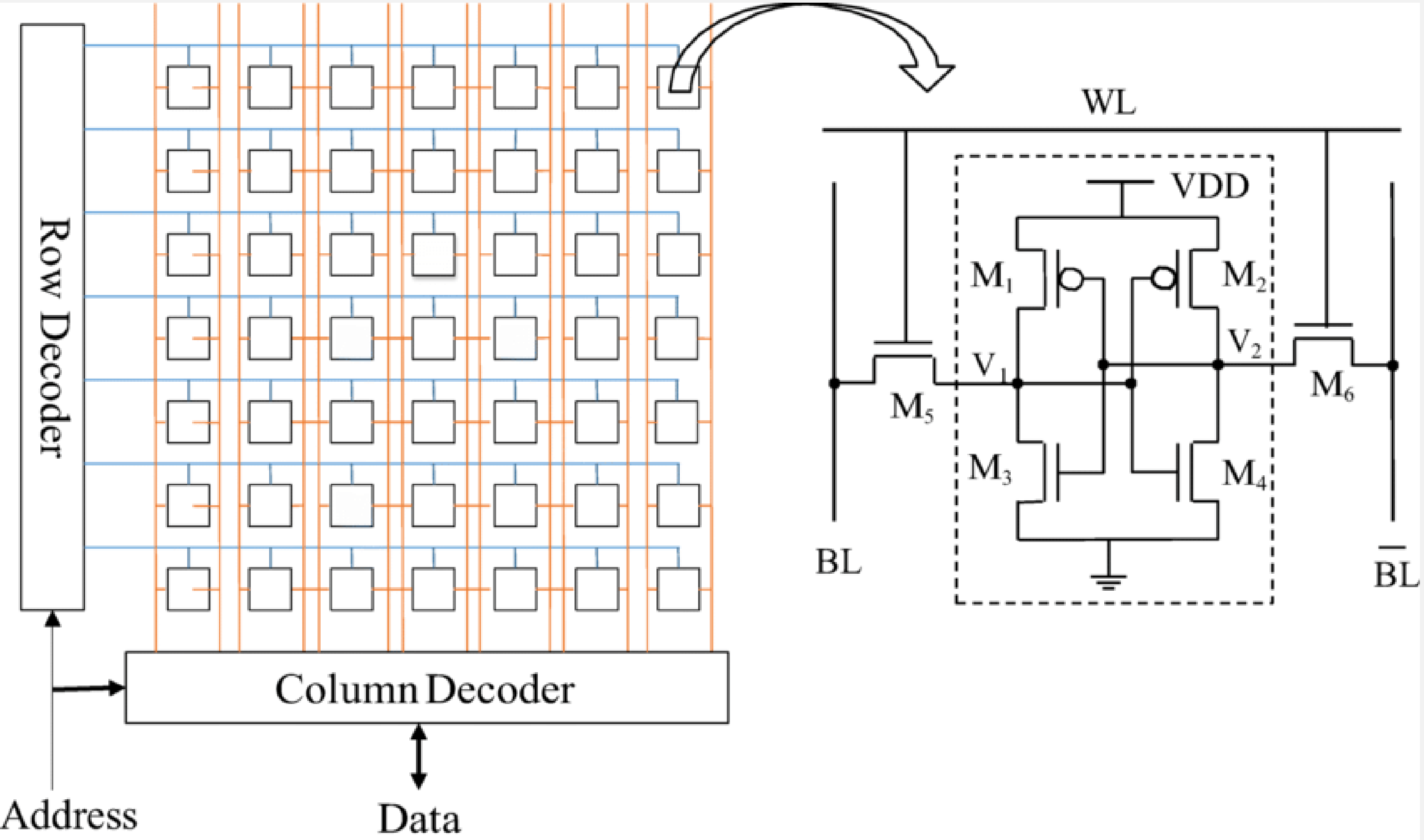
On-chip Memory: SRAM







SRAM Cell Structure



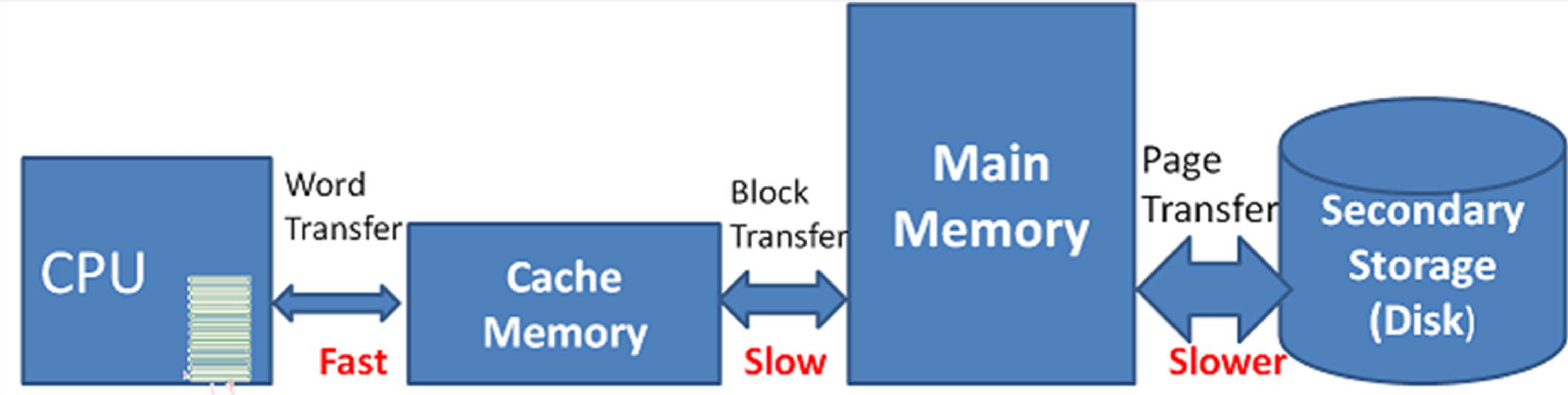
(a) SRAM array

(b) 6T SRAM cell

FAST and No need to refresh, but consume more transistors!

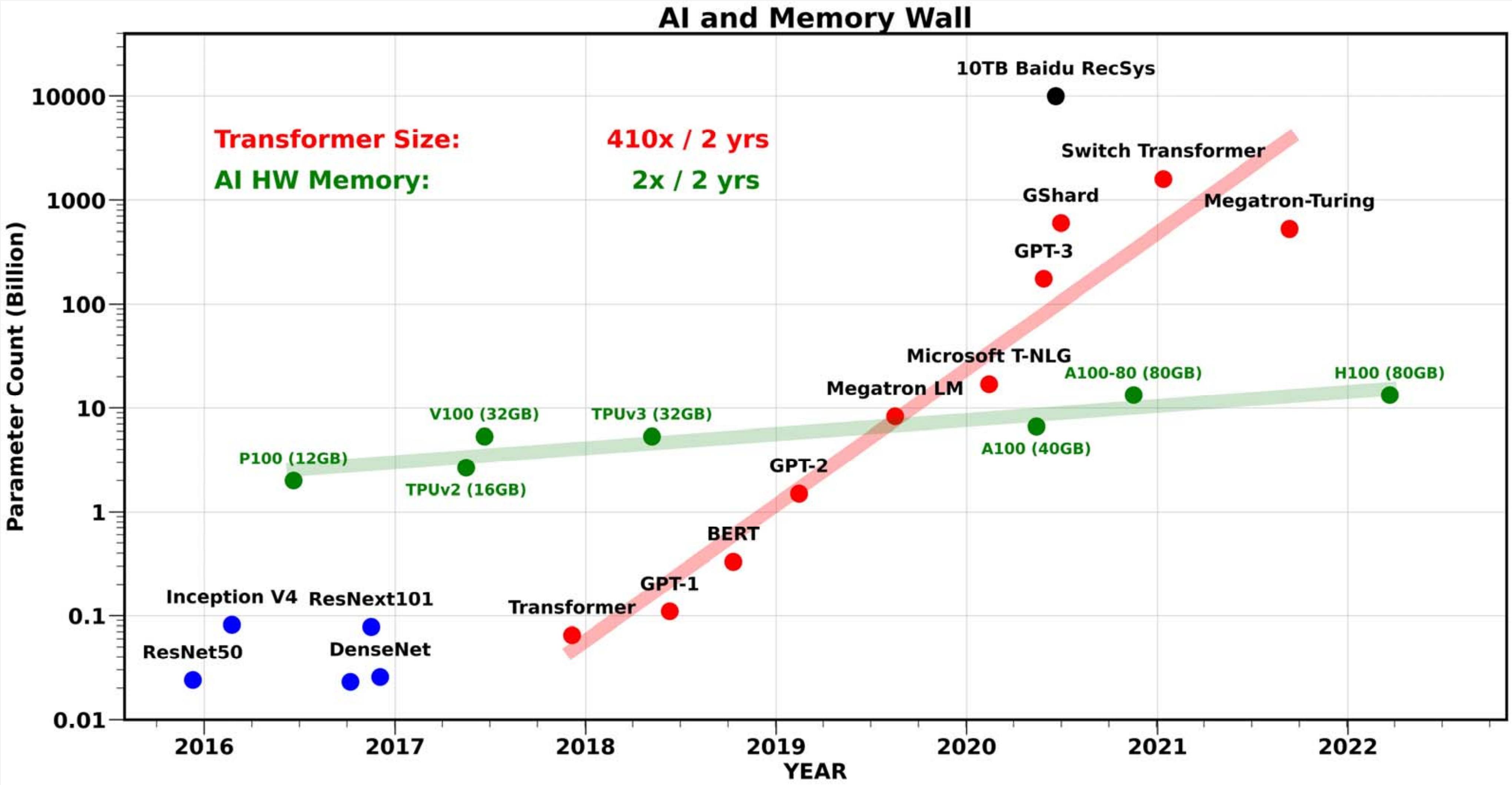
To Strive a Balance

上海科技大学
ShanghaiTech University

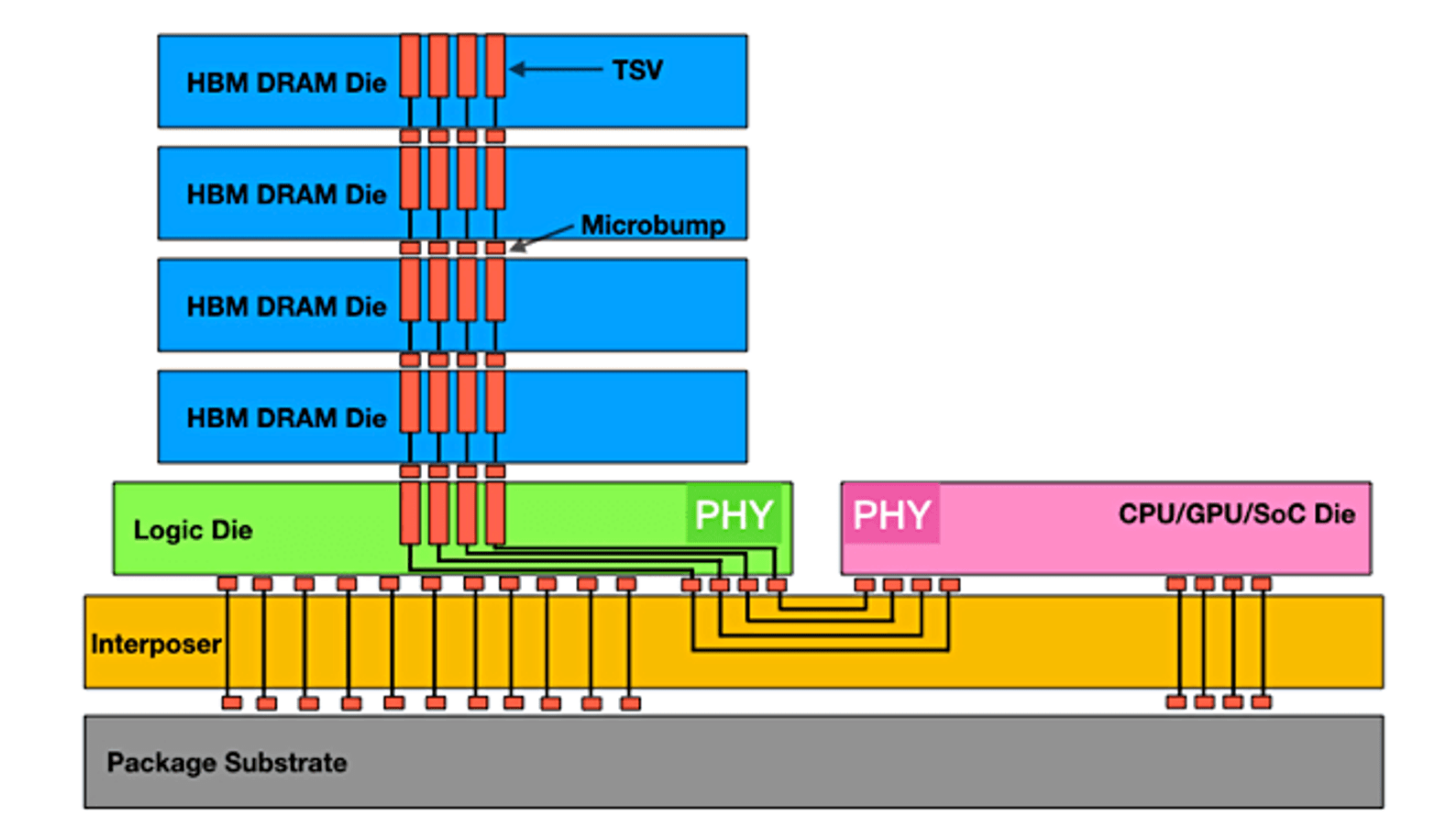


The Memory Wall

上海科技大学 ShanghaiTech University



High-bandwidth Memory (HBM)



High-bandwidth Memory (HBM)

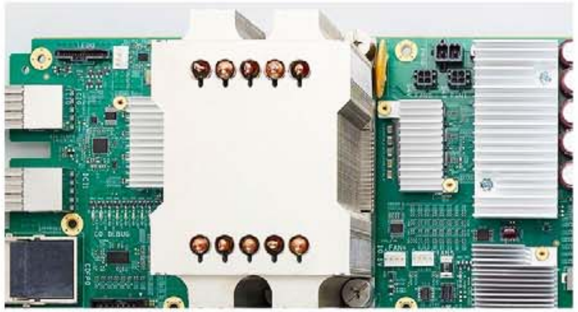
上海科技大学 ShanghaiTech University

GDDR vs LPDDR vs HBM

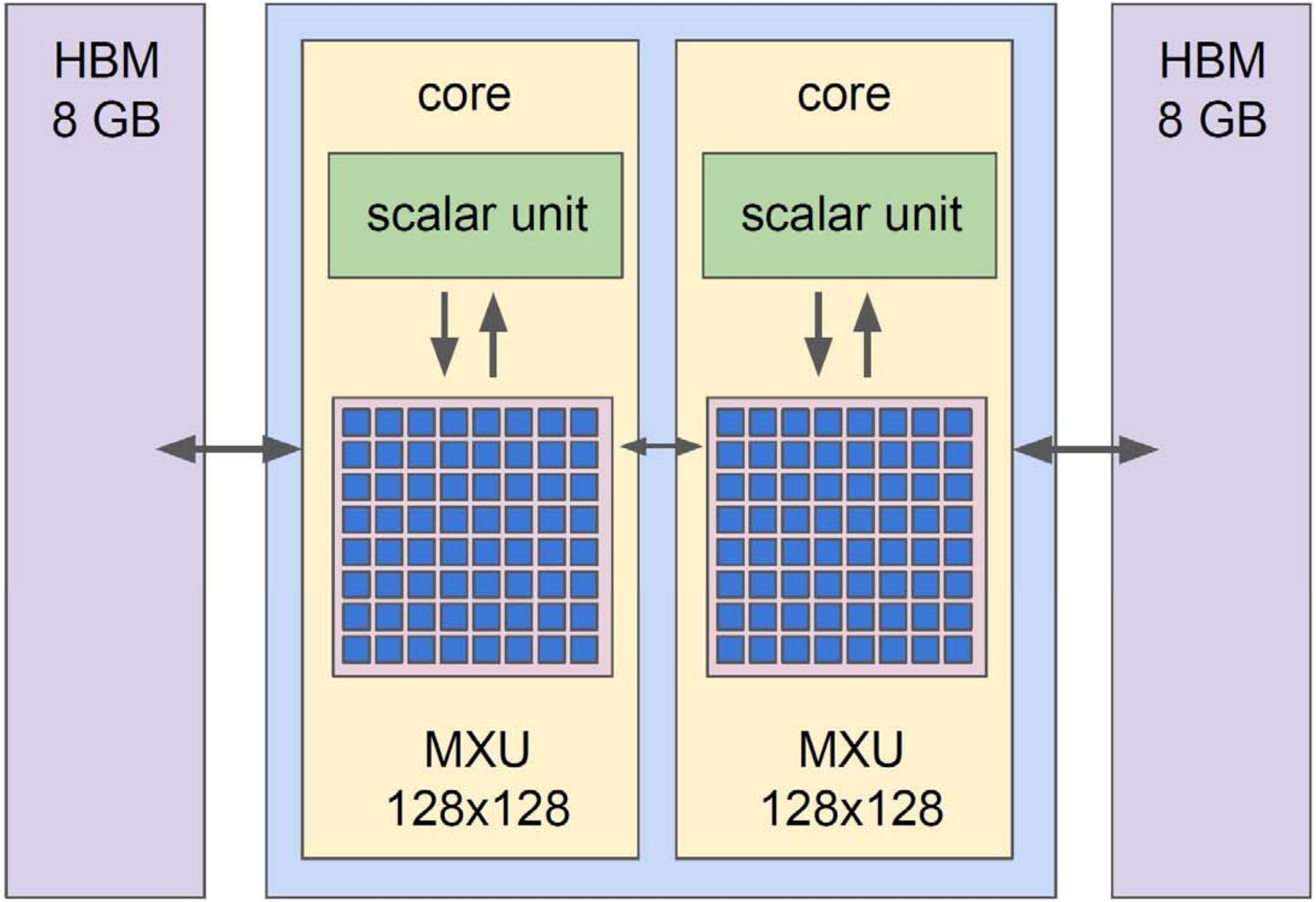
Property	HBM4	HBM3E	HBM3	HBM2E	GDDR7	GDDR6	LPDDR5X	DDR5
Max capacity per stack, chip or module	36-64GB	36GB	24GB	16GB	3GB	2GB	16GB	2048GB
Data Transfer Rate	6.4GT/s	8.8GT/s	6.4GT/s	3.6GT/s	32GT/s	24GT/s	9.6GT/s	8.4GT/s
Max stack	16	12	12	8	1*	1*	8	8-16
Interface Width	2048	1024	1024	1024	32	32	32-64	64
Signaling	Undisclosed	NRZ	NRZ	NRZ	PAM3	NRZ	NRZ	NRZ
I/O Voltage		1.1v	1.1v	1.2v	1.2v	1.2-1.35v	0.4v	1.1v
Bandwidth per stack, chip or module	1500-2000GB/s	1200GB/s	819GB/s	406GB/s	128GB/s	96GB/s	77GB/s	67GB/s

TPU V2

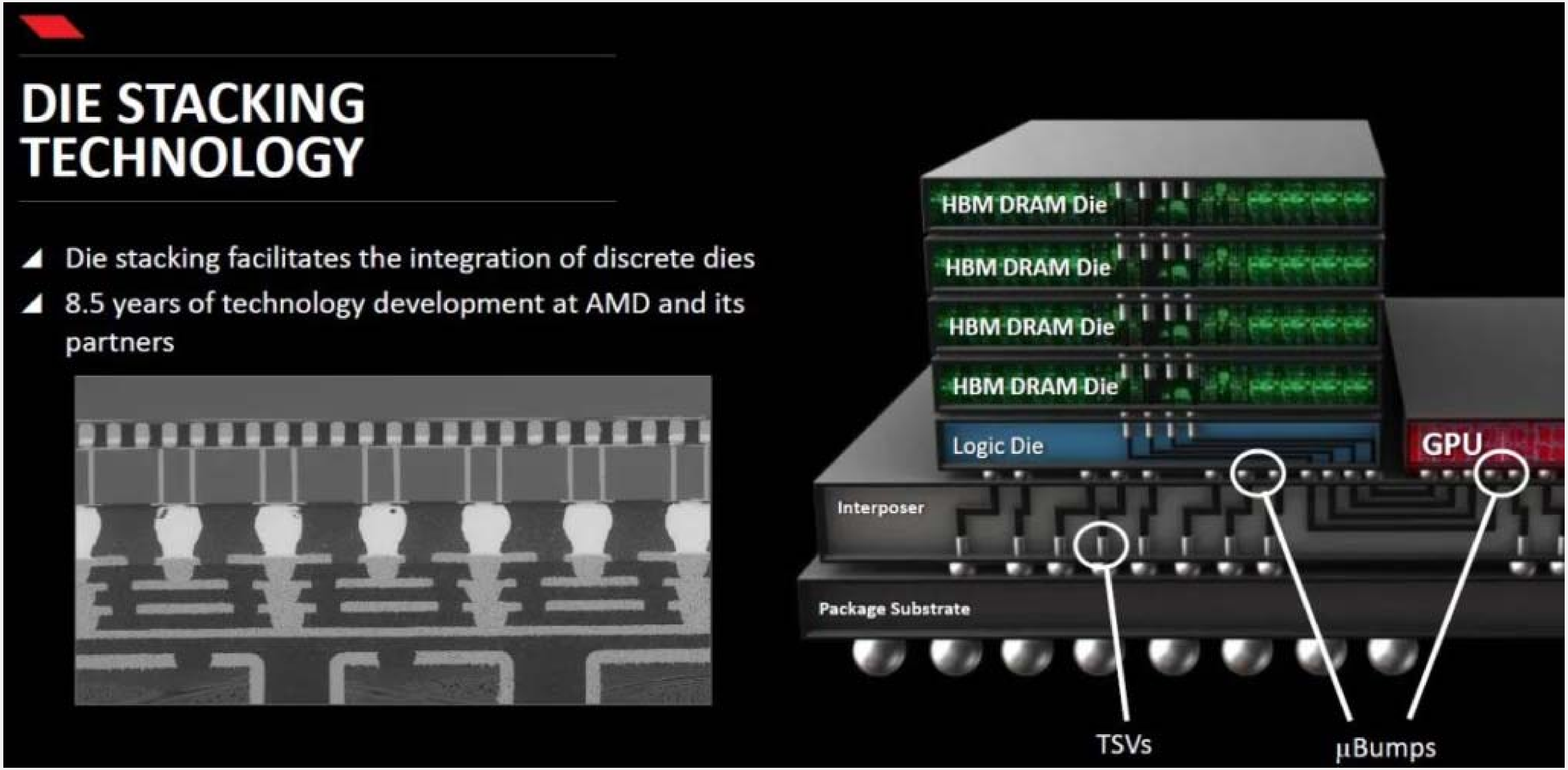
TPUv2 Chip



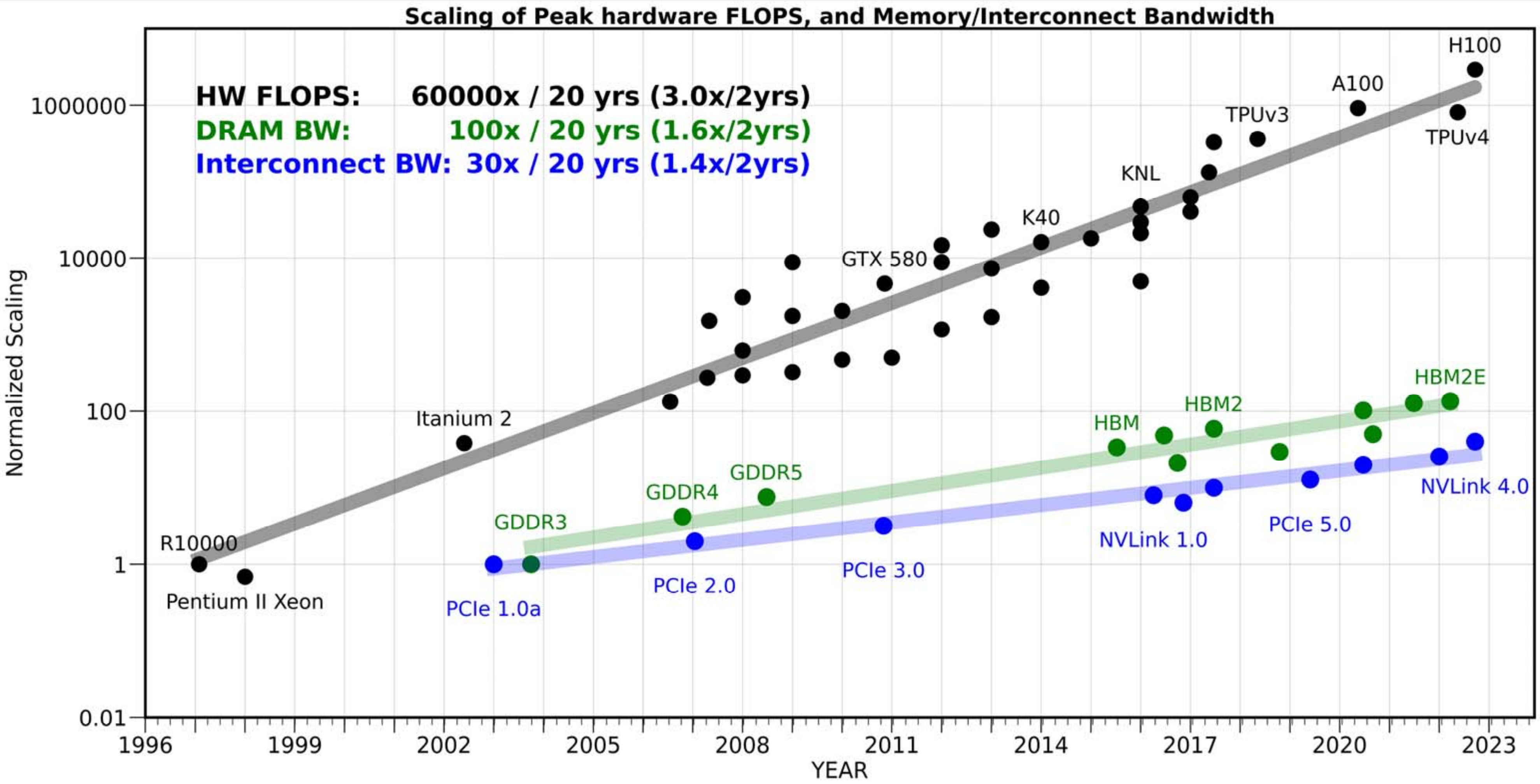
- 16 GB of HBM
- 600 GB/s mem BW
- Scalar unit: 32b float
- MXU: 32b float accumulation but reduced precision for multipliers
- 45 TFLOPS



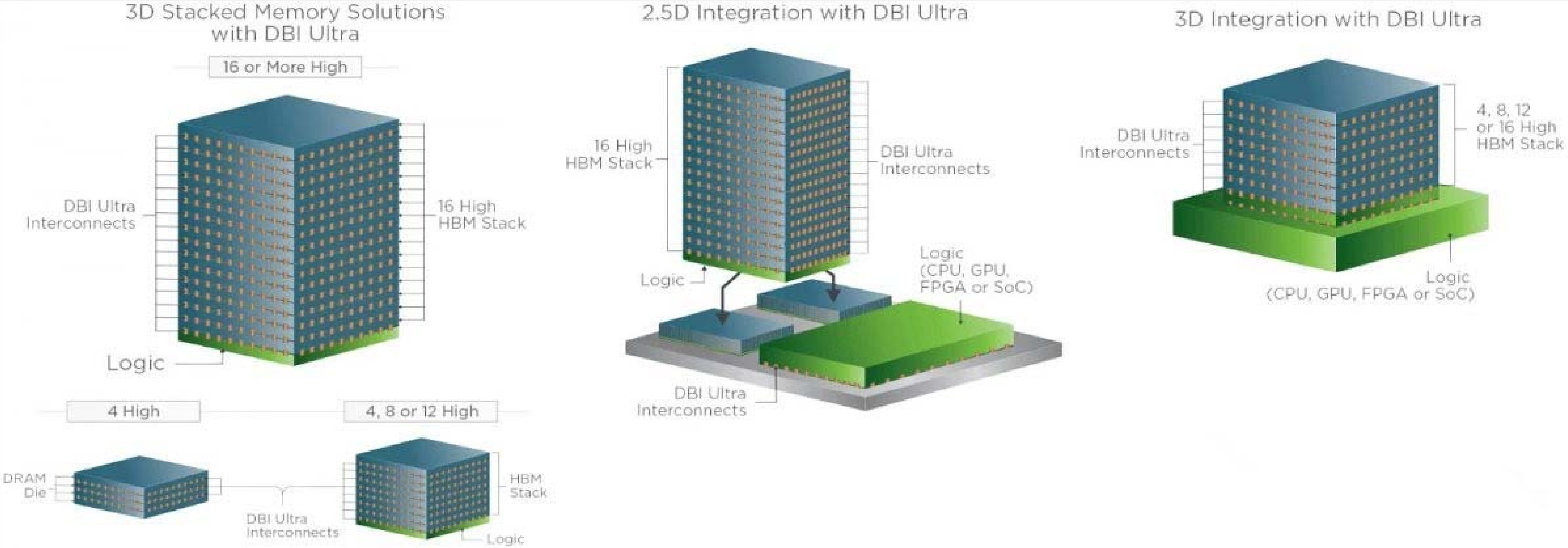
HBM in AMD GPU



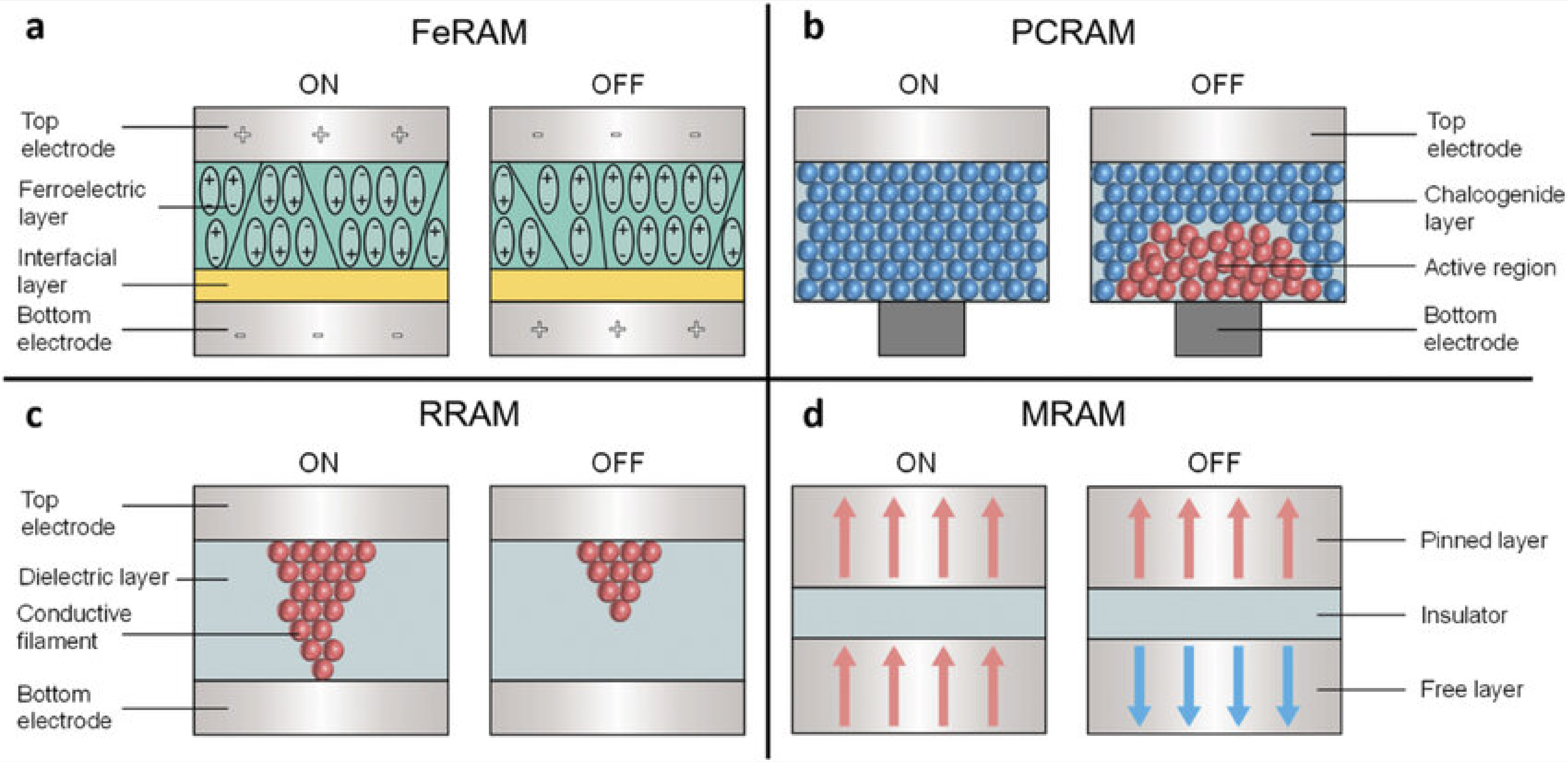
Memory Wall Continues



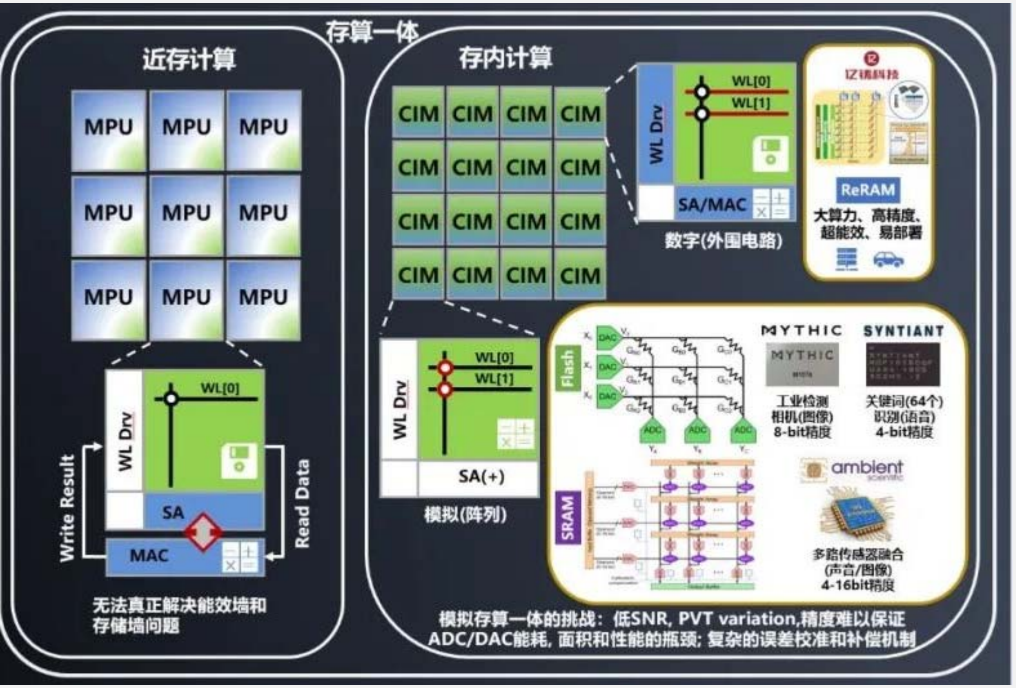
3D integration



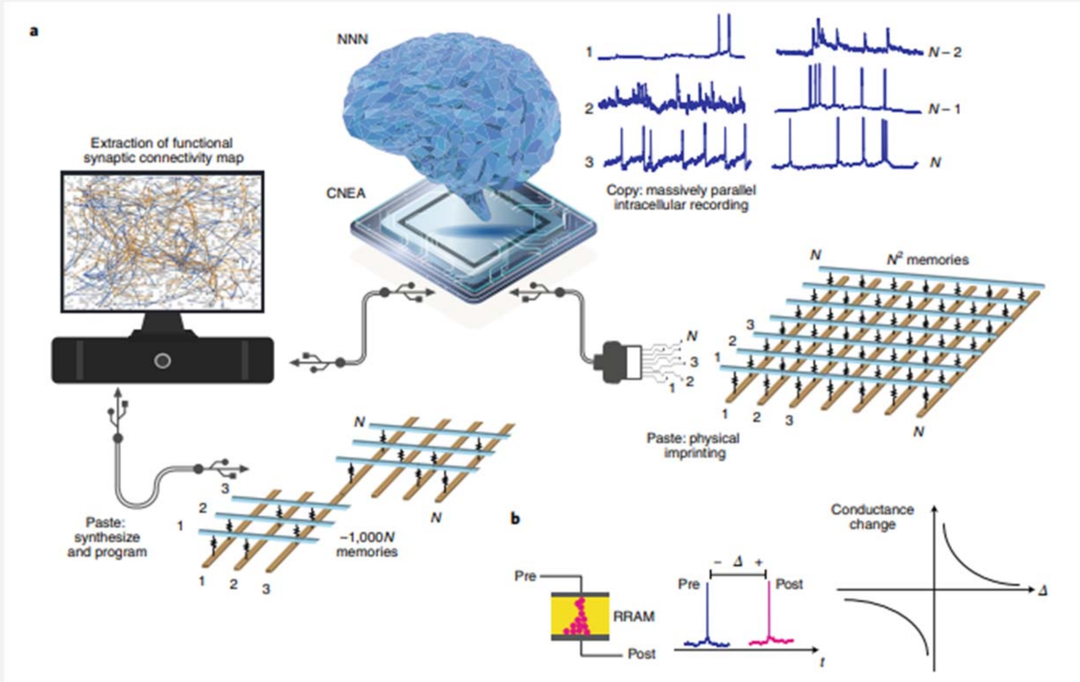
Other Memory Technologies



Other Computation Architectures



In Memory Computing

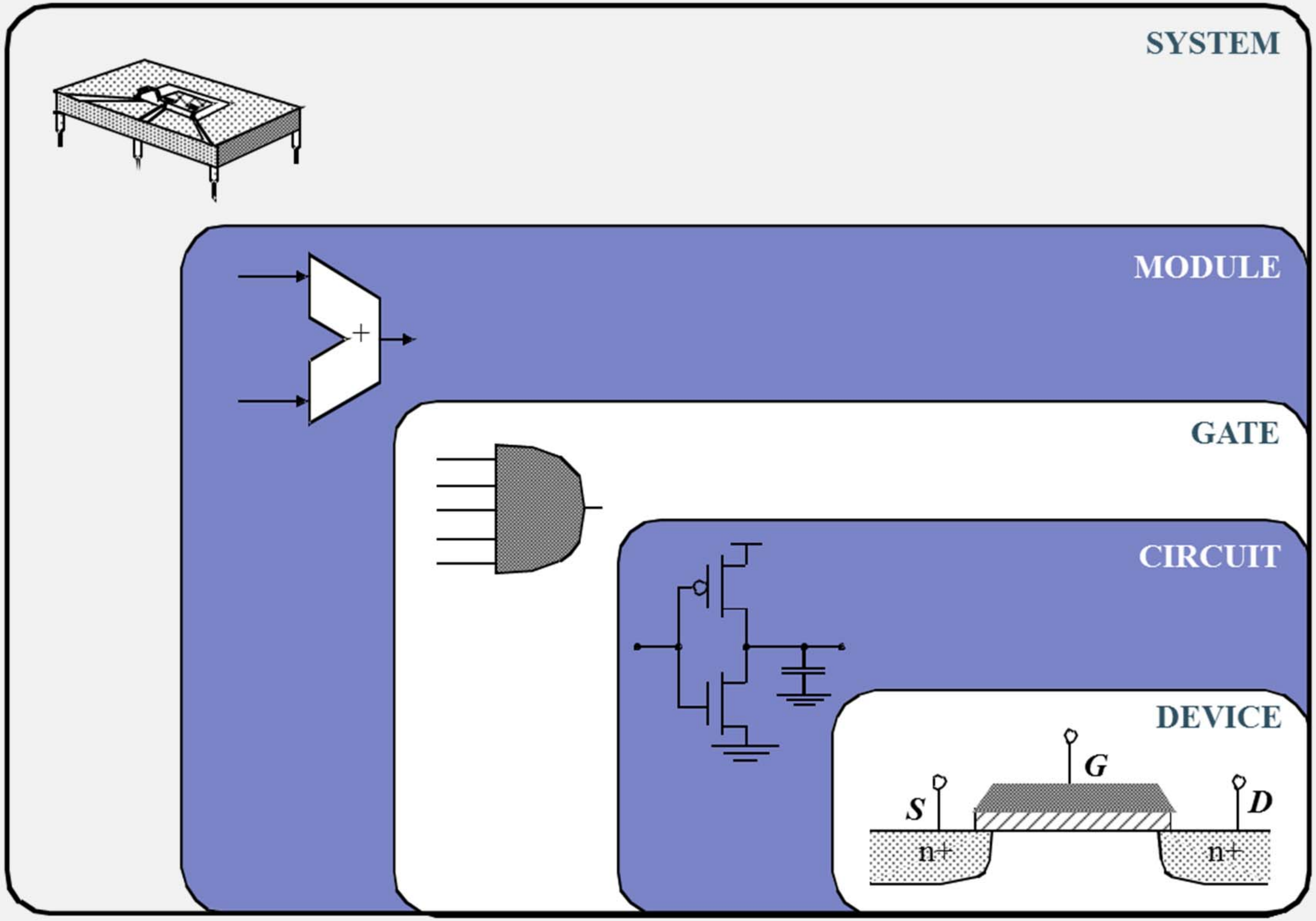


Neuromorphic

Logic

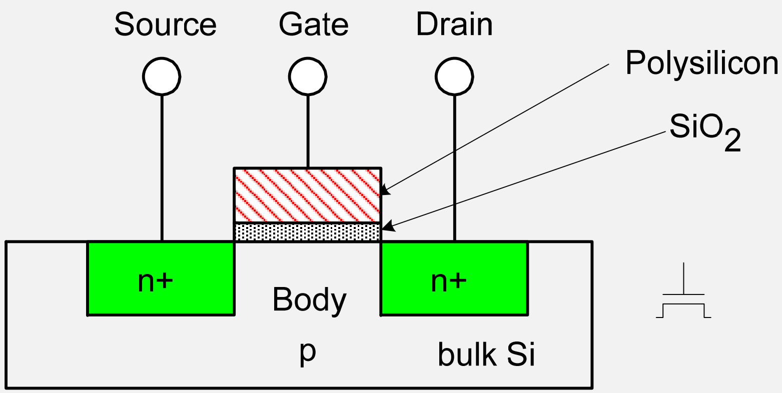
A component that do computation.

Abstraction Levels



nMOS Transistor

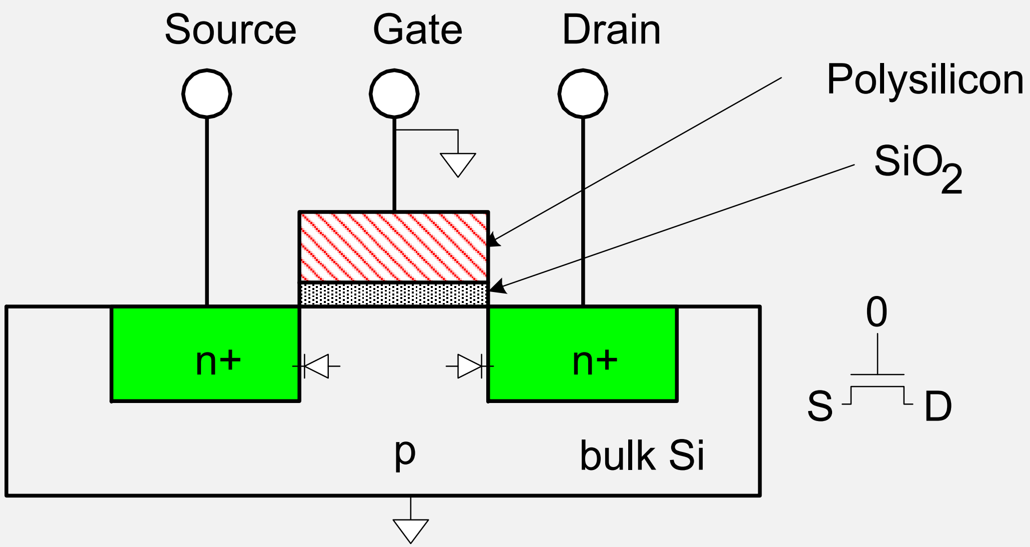
- Four terminals: gate, source, drain, body
- Gate–oxide–body stack looks like a capacitor
  - Gate and body are conductors
  - SiO_2 (oxide) is a very good insulator
  - Called metal–oxide–semiconductor (MOS) capacitor
  - Even though gate is no longer made of metal*



* Metal gates are returning today!

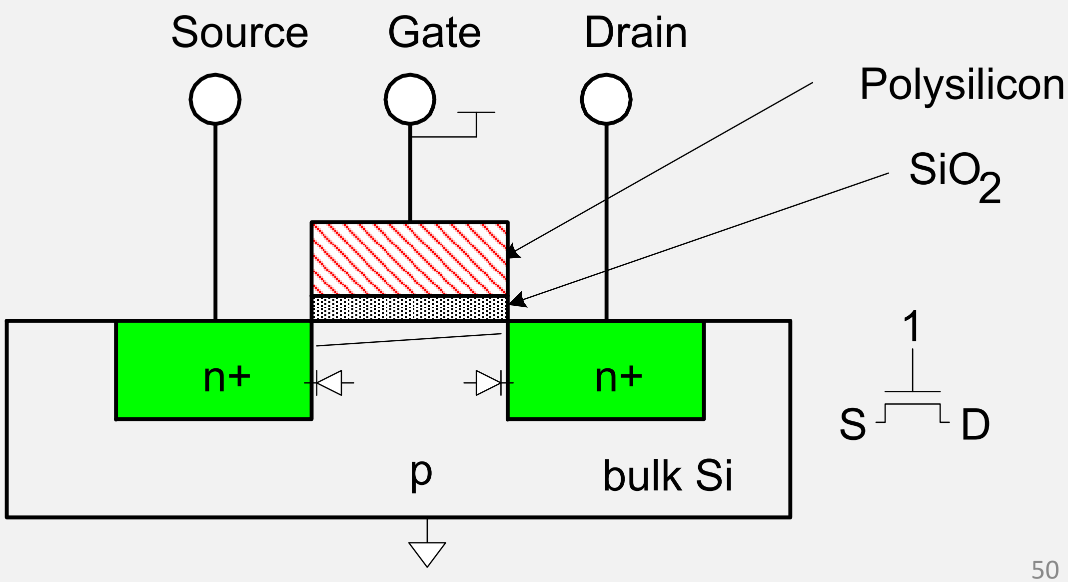
nMOS Operation

- Body is usually tied to ground (0 V)
- When the gate is at a low voltage:
  - P-type body is at low voltage
  - Source-body and drain-body diodes are OFF
  - No current flows, transistor is OFF



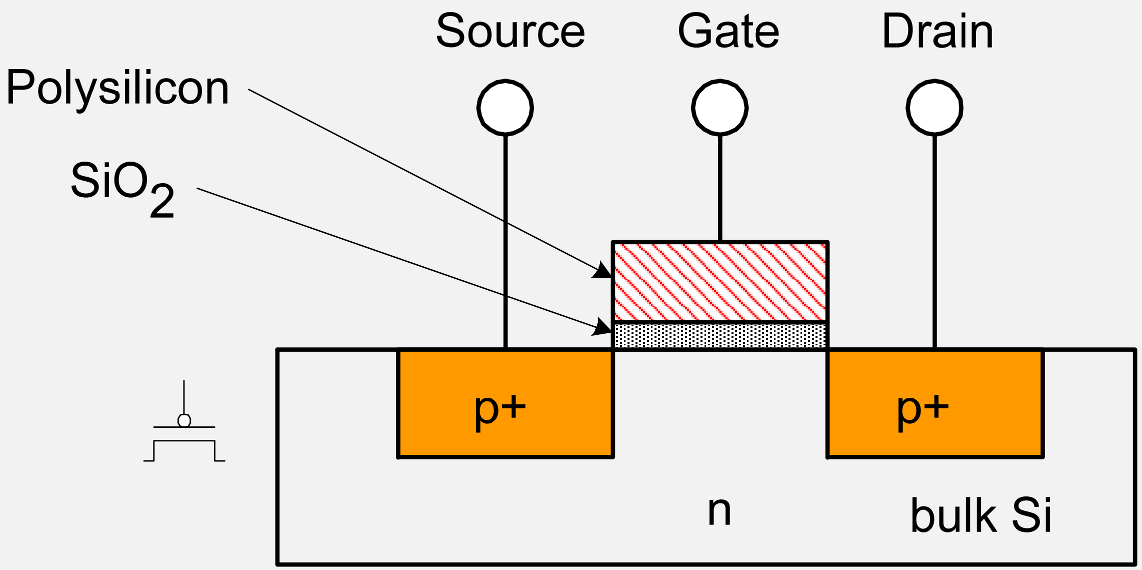
nMOS Operation

- When the gate is at a high voltage:
  - Positive charge on gate of MOS capacitor
  - Negative charge attracted to body
  - Inverts a channel under gate to n-type
  - Now current can flow through n-type silicon from source through channel to drain, transistor is ON



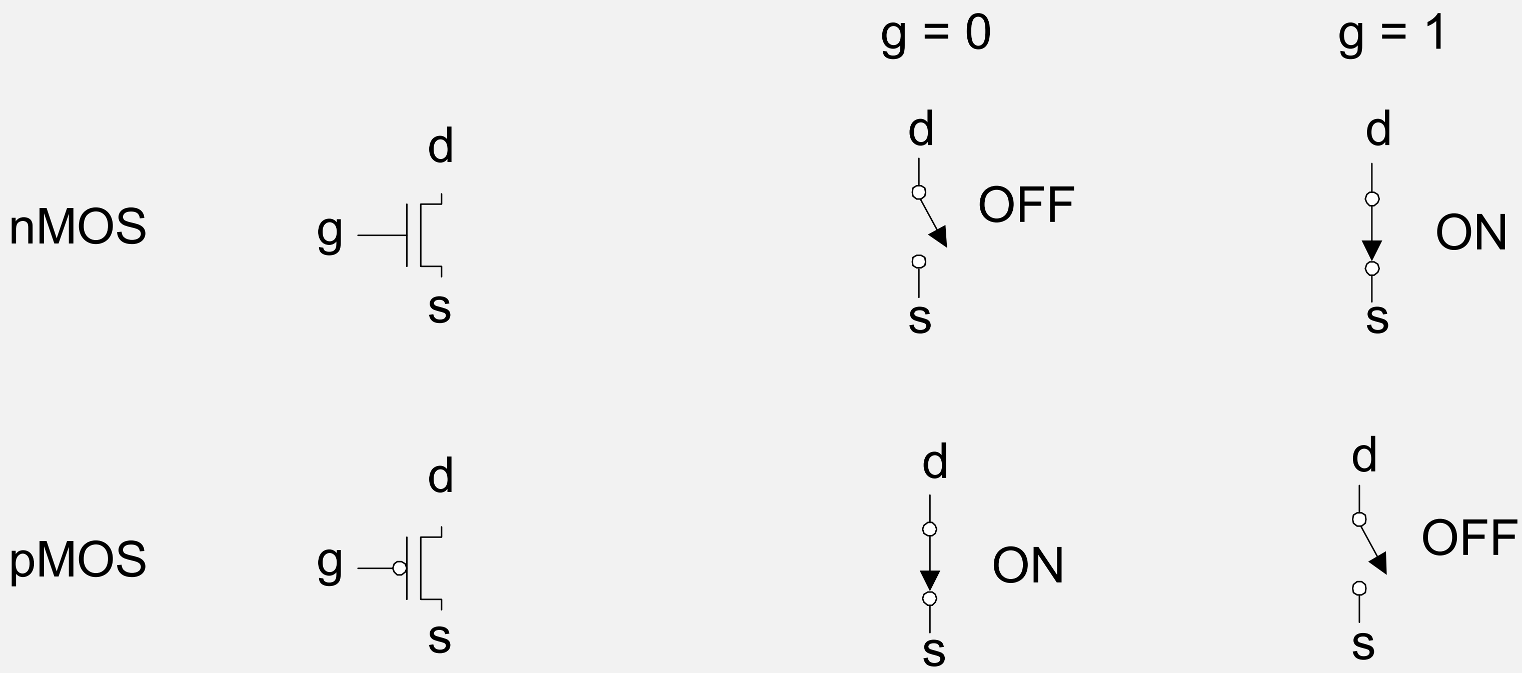
pMOS Transistor

- Similar, but doping and voltages reversed
  - Body tied to high voltage (V_{DD})
  - Gate low: transistor ON
  - Gate high: transistor OFF
  - Bubble indicates inverted behavior



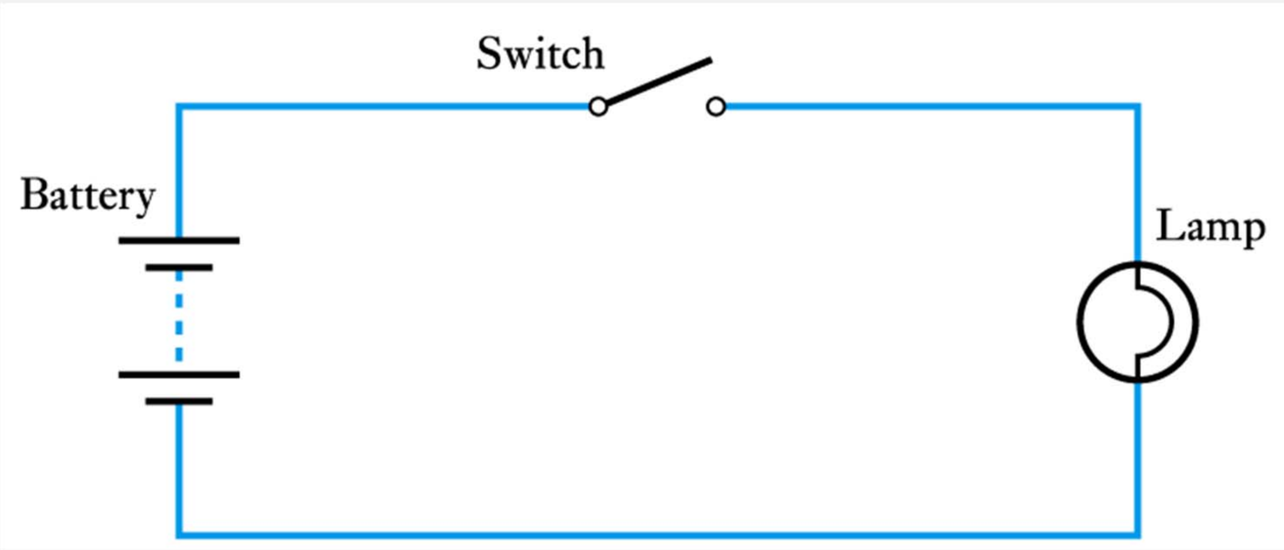
Transistors as Switches

- We can view MOS transistors as electrically controlled switches
- Voltage at gate controls path from source to drain



Binary Quantities

- A binary quantity is one that can take only 2 states



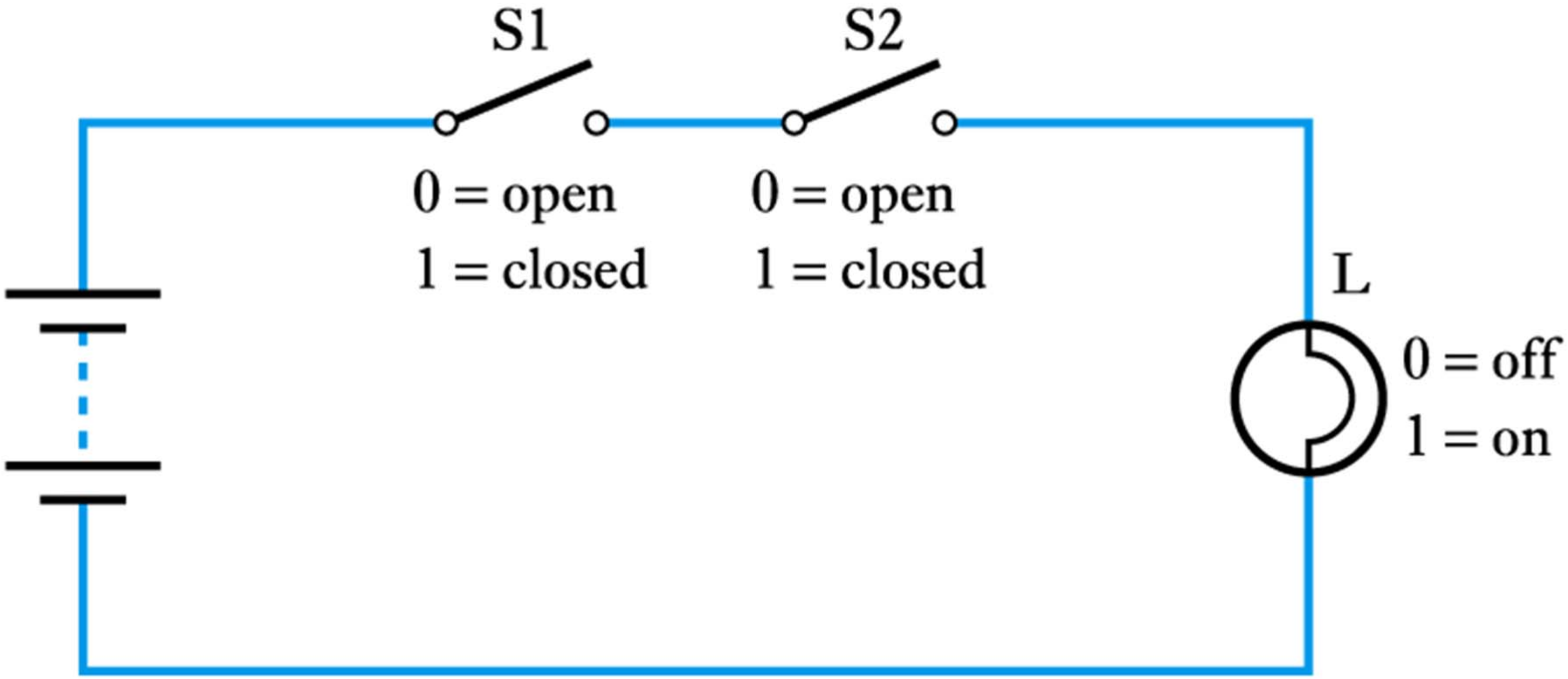
A simple binary arrangement

S	*L*
OPEN	OFF
CLOSED	ON

S	*L*
0	0
1	1

A truth table

Two Switches in Series



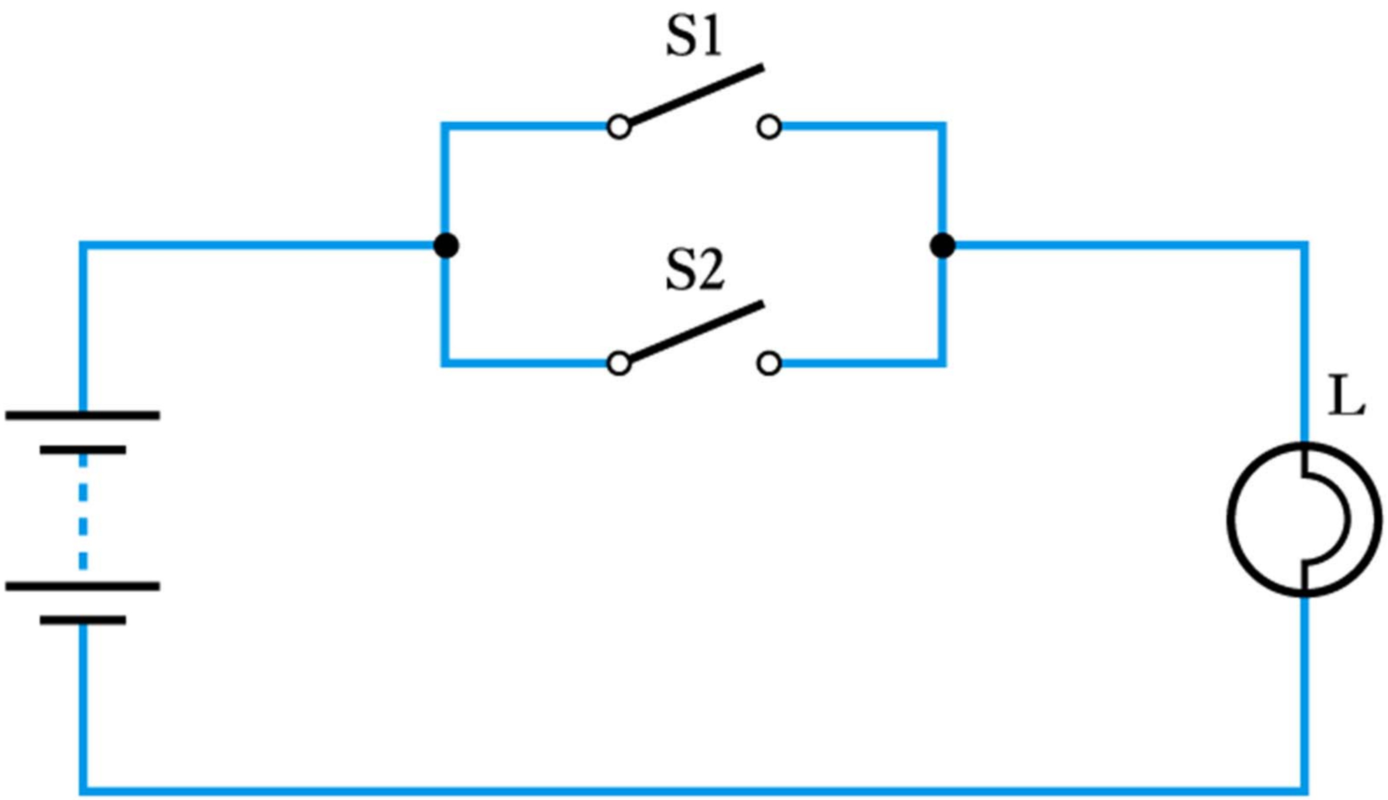
(a) Circuit

S1	S2	L
0	0	0
0	1	0
1	0	0
1	1	1

(b) Truth table

$$L = S1 \text{ AND } S2$$

Two Switches in Parallel



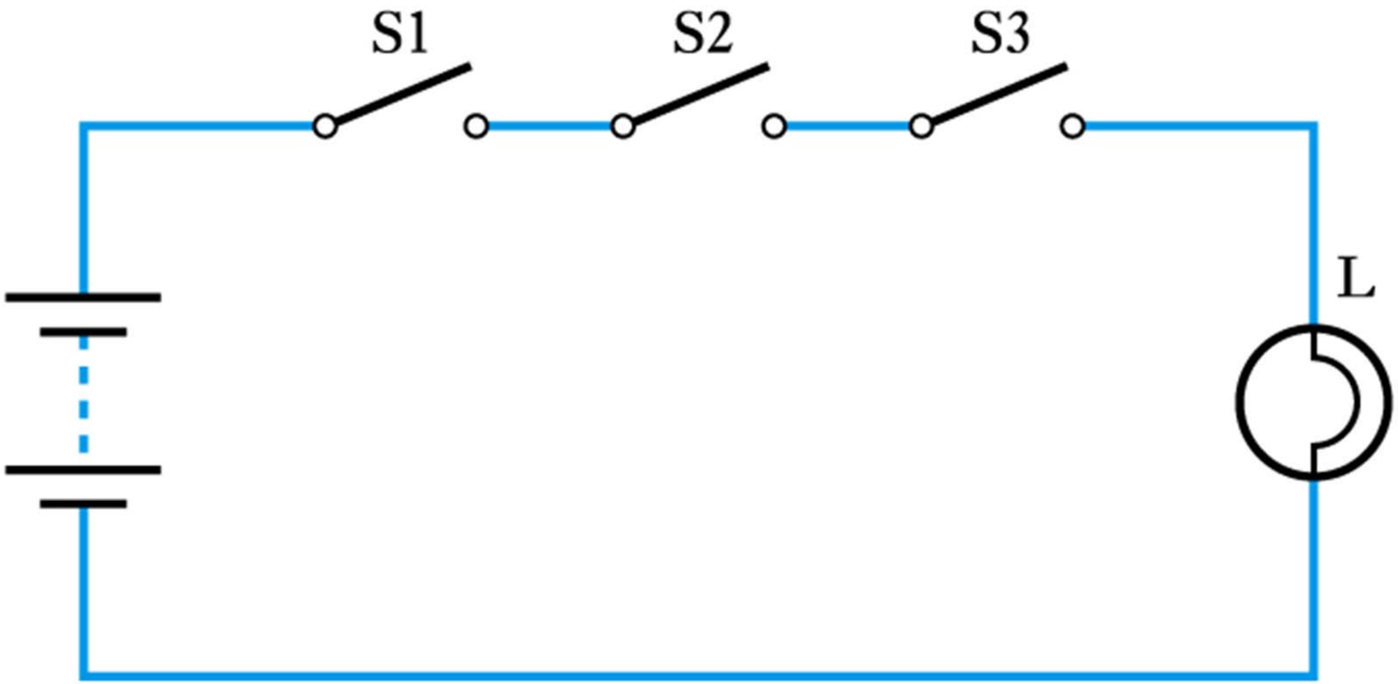
(a) Circuit

S1	S2	L
0	0	0
0	1	1
1	0	1
1	1	1

(b) Truth table

$$L = S1 \text{ OR } S2$$

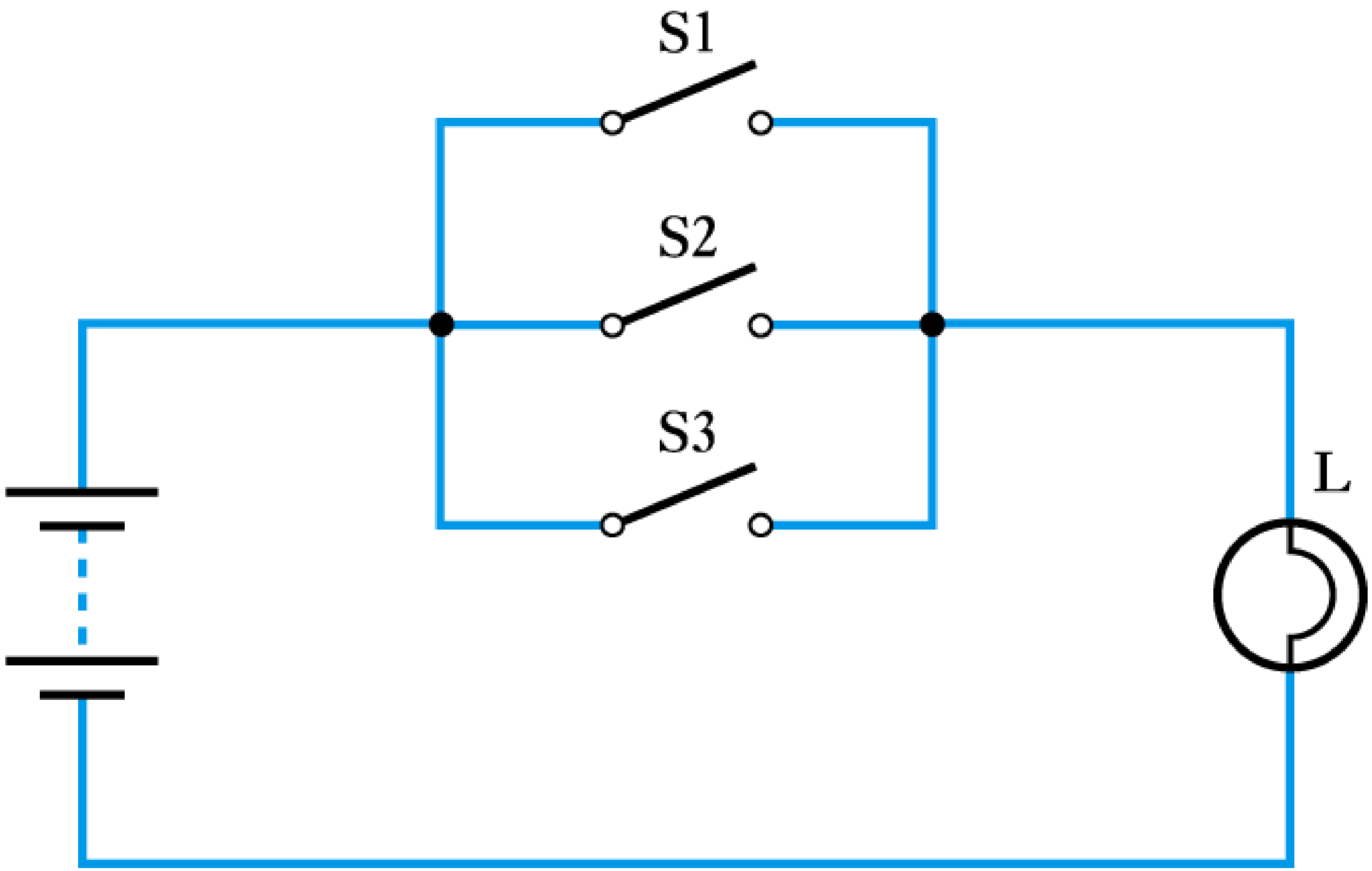
Three Switches in Series



S1	S2	S3	L
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

$L = S1$ AND $S2$ AND $S3$

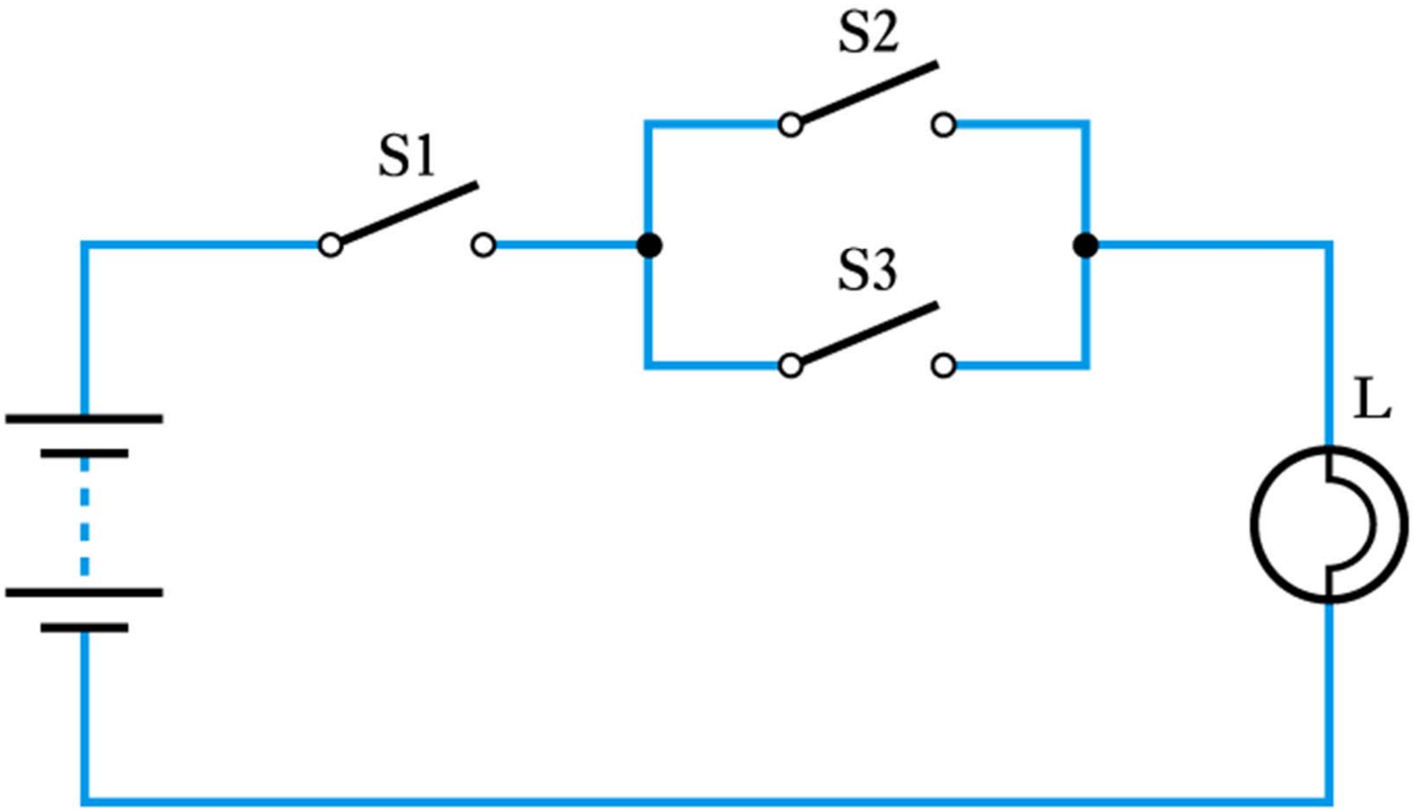
Three Switches in Parallel



S1	S2	S3	L
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$L = S1$ OR $S2$ OR $S3$

A Series/Parallel Arrangement

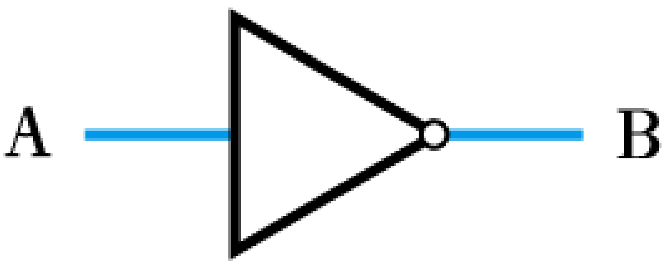


S1	S2	S3	L
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$L = S1$ AND ($S2$ OR $S3$)

The NOT Gate

➢The NOT gate (or inverter)



(a) Circuit symbol

A	B
0	1
1	0

(b) Truth table

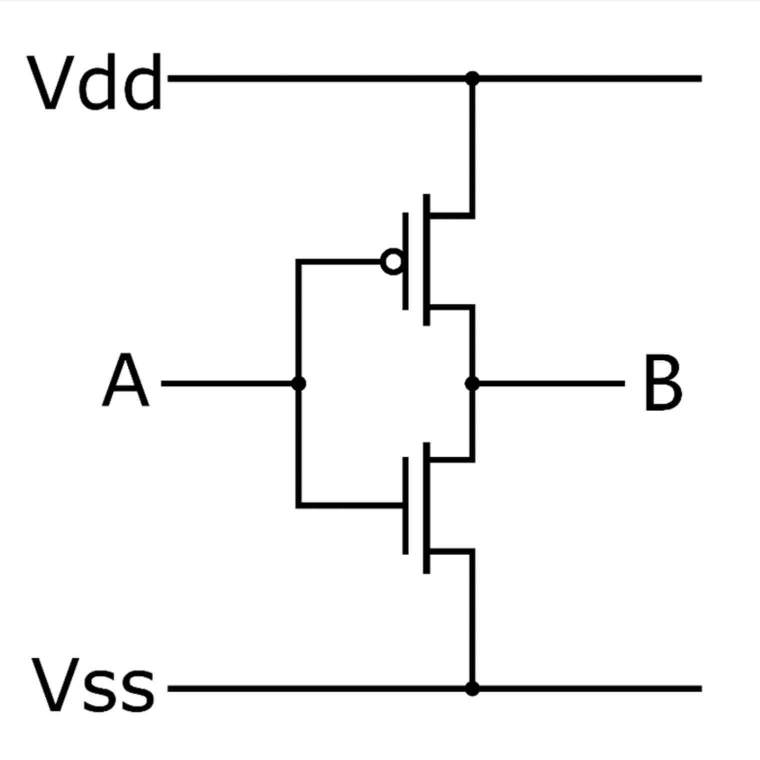
$$B = \overline{A}$$

(c) Boolean expression

❖B is TRUE if A is FALSE

Implementation of NOT Gate

➢ Transistor-level implementation of a NOT gate

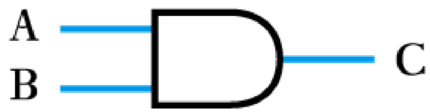


$$B = \overline{A}$$

- If A=1
  - The nMOS (bottom one) is on
  - The pMOS (upper one) is off
  - B=0
- If A=0
  - The nMOS (bottom one) is off
  - The pMOS (upper one) is on
  - B=1

The AND Gate

- The AND gate



(a) Circuit symbol

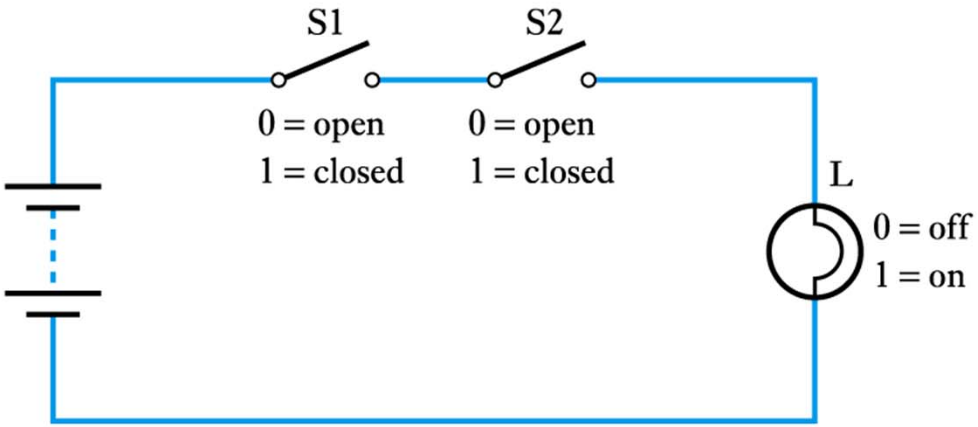
A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

(b) Truth table

$$C = A \cdot B$$

(c) Boolean expression

- C is TRUE only if A is TRUE and B is TRUE



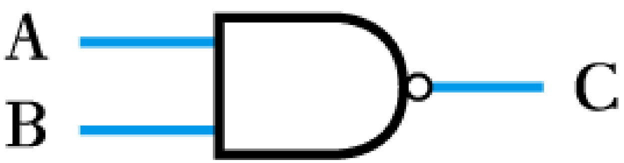
(a) Circuit

S1	S2	L
0	0	0
0	1	0
1	0	0
1	1	1

(b) Truth table

The NAND Gate

- The NAND Gate



(a) Circuit symbol

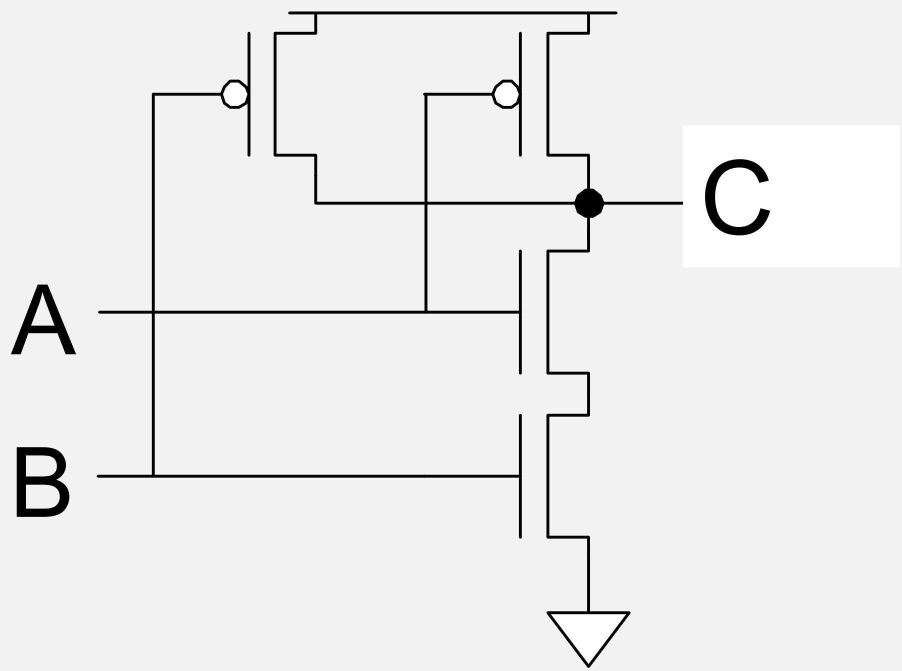
A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

(b) Truth table

$$C = \overline{A \cdot B}$$

(c) Boolean expression

The NAND Gate Cont'd



A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

The OR Gate

- The OR gate

(a) Circuit symbol

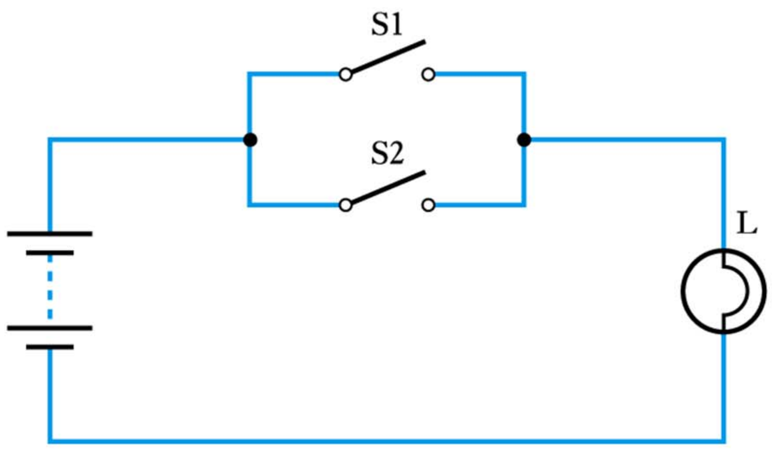
A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

(b) Truth table

$$C = A + B$$

(c) Boolean expression

  - C is TRUE if A is TRUE or B is TRUE or both



(a) Circuit

S1	S2	L
0	0	0
0	1	1
1	0	1
1	1	1

(b) Truth table

The NOR Gate

➢The NOR gate

(a) Circuit symbol

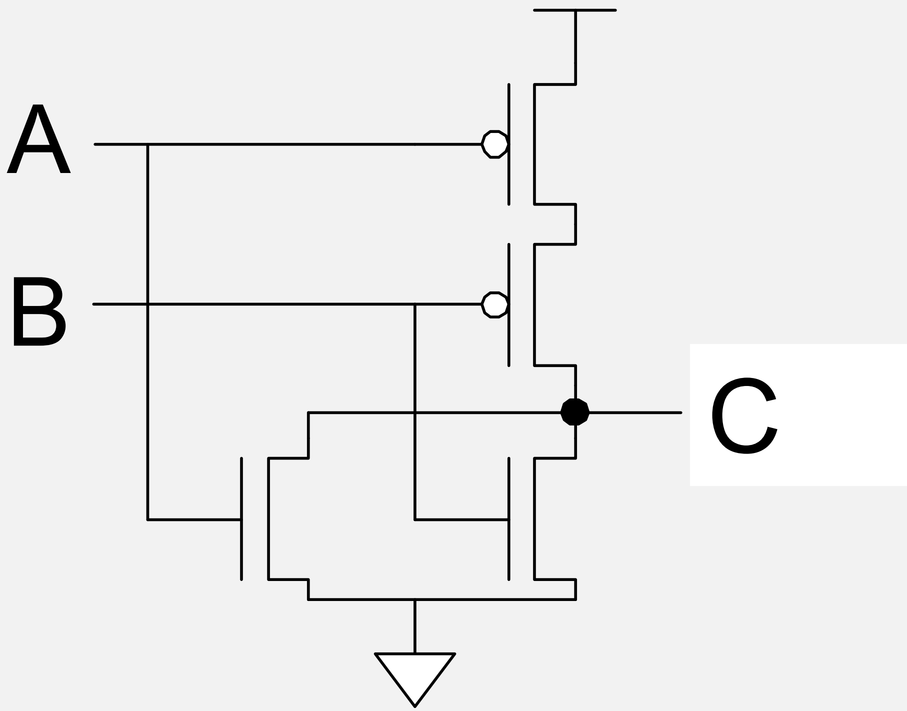
A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

(b) Truth table

$$C = \overline{A + B}$$

(c) Boolean expression

The NOR Gate Cont'd

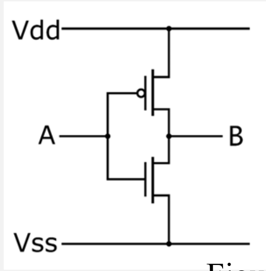


A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

- (1) Fill in the highlighted blanks in Figure 1.

Figure 1

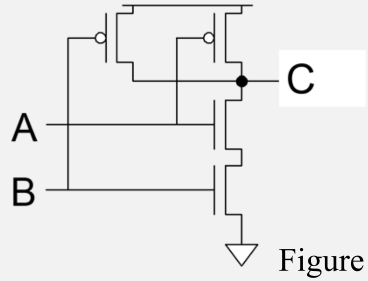
- (2) Fill in the highlighted blanks in Figure 2.



A	B
0	
1	

Figure 2

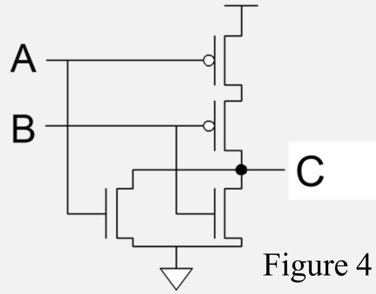
- (3) Fill in the highlighted blanks in Figure 3.



A	B	C
0	0	
0	1	
1	0	
1	1	

Figure 3

- (4) Fill in the highlighted blanks in Figure 4.



A	B	C
0	0	
0	1	
1	0	
1	1	

Figure 4

The Exclusive OR Gate

➢The Exclusive OR gate

(a) Circuit symbol

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

(b) Truth table

$$C = A \oplus B$$

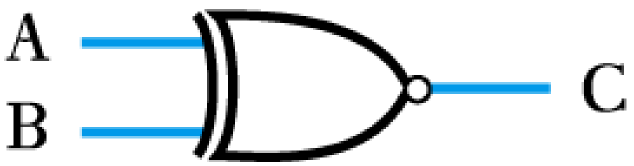
(c) Boolean expression

❖C is TRUE if A is TRUE or B is TRUE but not both

❖Or, C is TRUE if A and B are different (exclusive)

The Exclusive NOR Gate

- The Exclusive NOR gate



(a) Circuit symbol

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

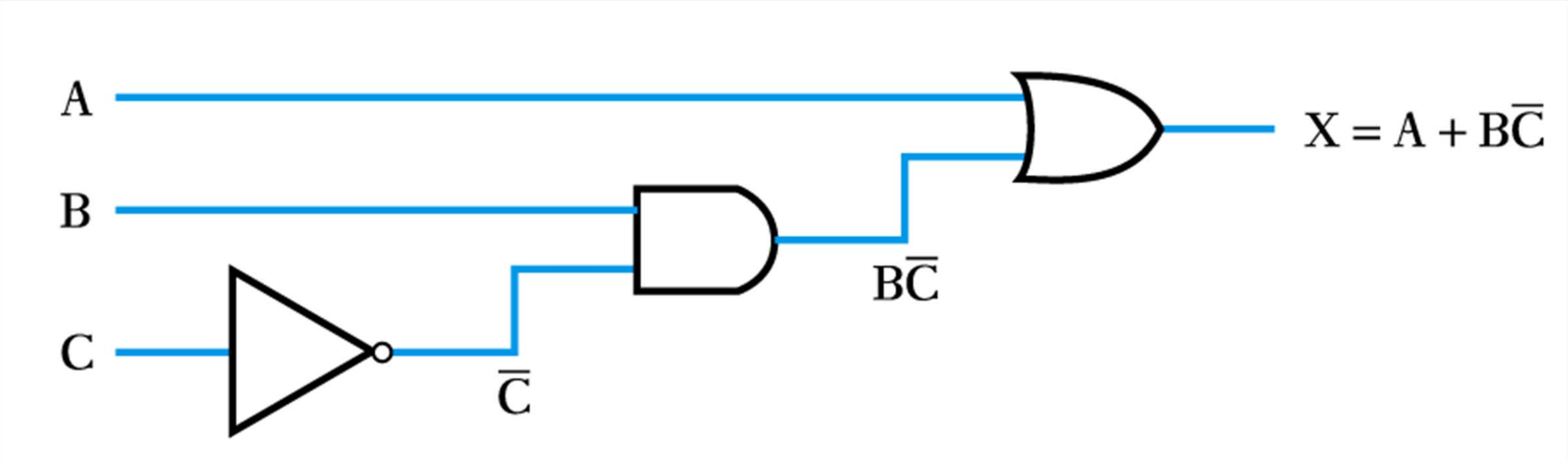
(b) Truth table

$$C = \overline{A \oplus B}$$

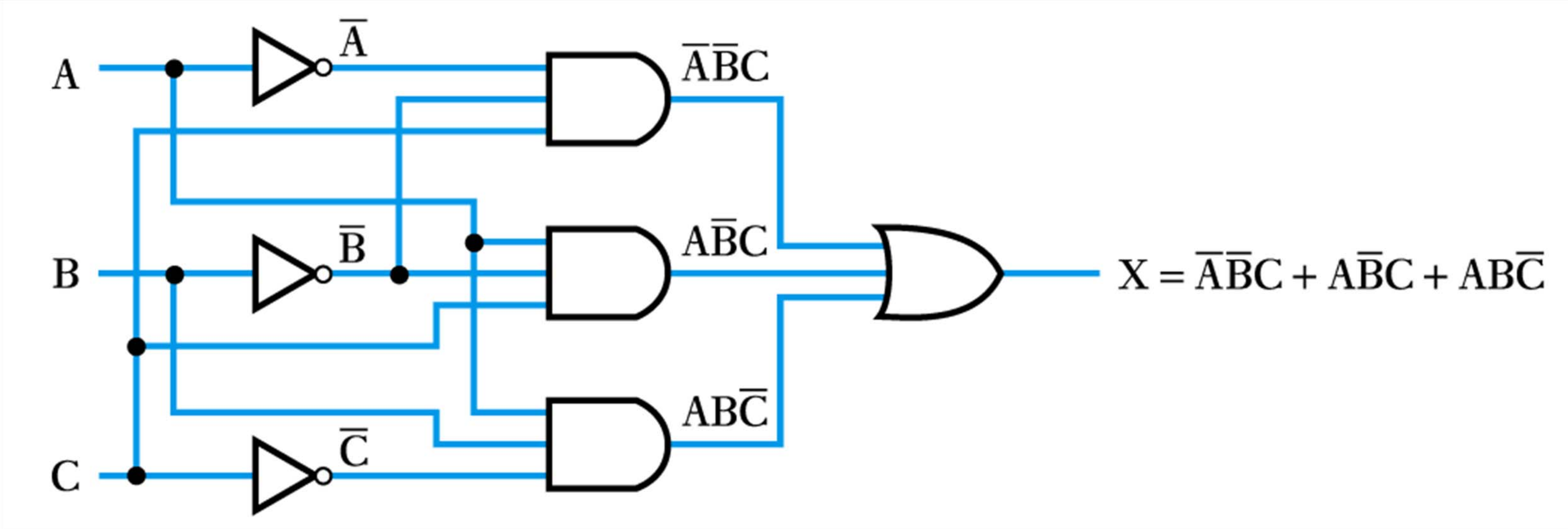
(c) Boolean expression

More Complicated function

- Implement the function $X = A + B\overline{C}$

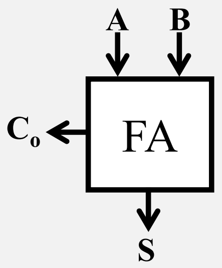


- The logic function is $X = \overline{A}\,\overline{B}C + A\overline{B}C + AB\overline{C}$



Single-Bit Addition

Half Adder

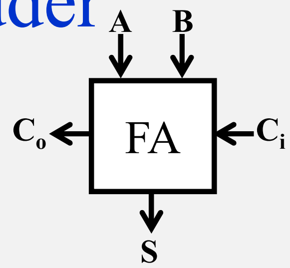


A	B	C_o	S
0	0		
0	1		
1	0		
1	1		

$S =$

$C_o =$

Full Adder

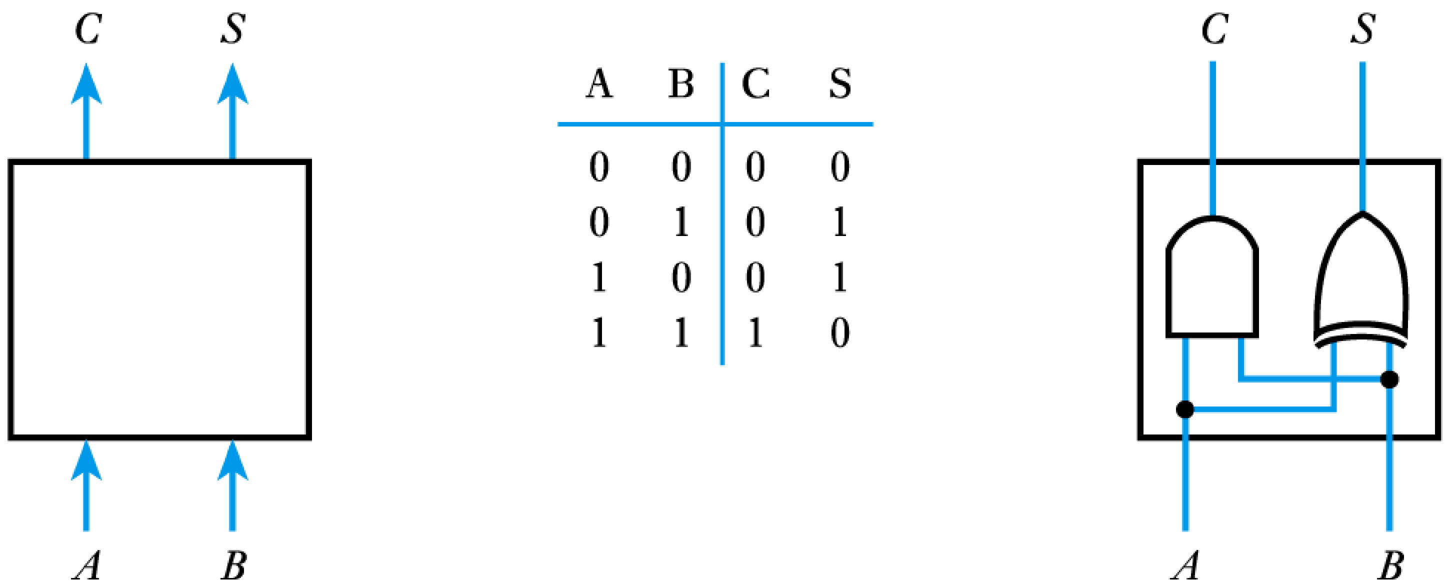


A	B	C_i	C_o	S
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

Half Adder

➢Gate implementation of a half adder

$$S = \overline{A}\cdot B + A\cdot \overline{B} = (A \oplus B) \qquad C_o = A\cdot B$$

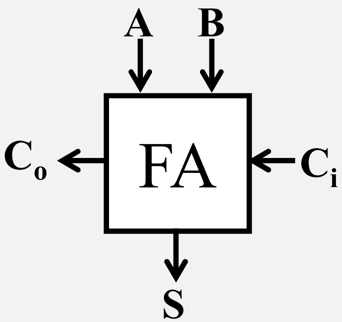


A	B	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

(a) Block diagram (b) Truth table (c) Circuit diagram

Full Adder

- Gate implementation of a full adder



A	B	C_i	C_o	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

$$\begin{aligned} S &= A\cdot B\cdot C_i + A\cdot \overline{B}\cdot \overline{C}_i + \overline{A}\cdot B\cdot \overline{C}_i + \overline{A}\cdot \overline{B}\cdot C_i \\ &= (A\cdot B + \overline{A}\cdot \overline{B})\cdot C_i + (A\cdot \overline{B} + \overline{A}\cdot B)\cdot \overline{C}_i \\ &= \overline{A\oplus B}\cdot C_i + A\oplus B\cdot \overline{C}_i \\ &= A\oplus B\oplus C_i \end{aligned}$$

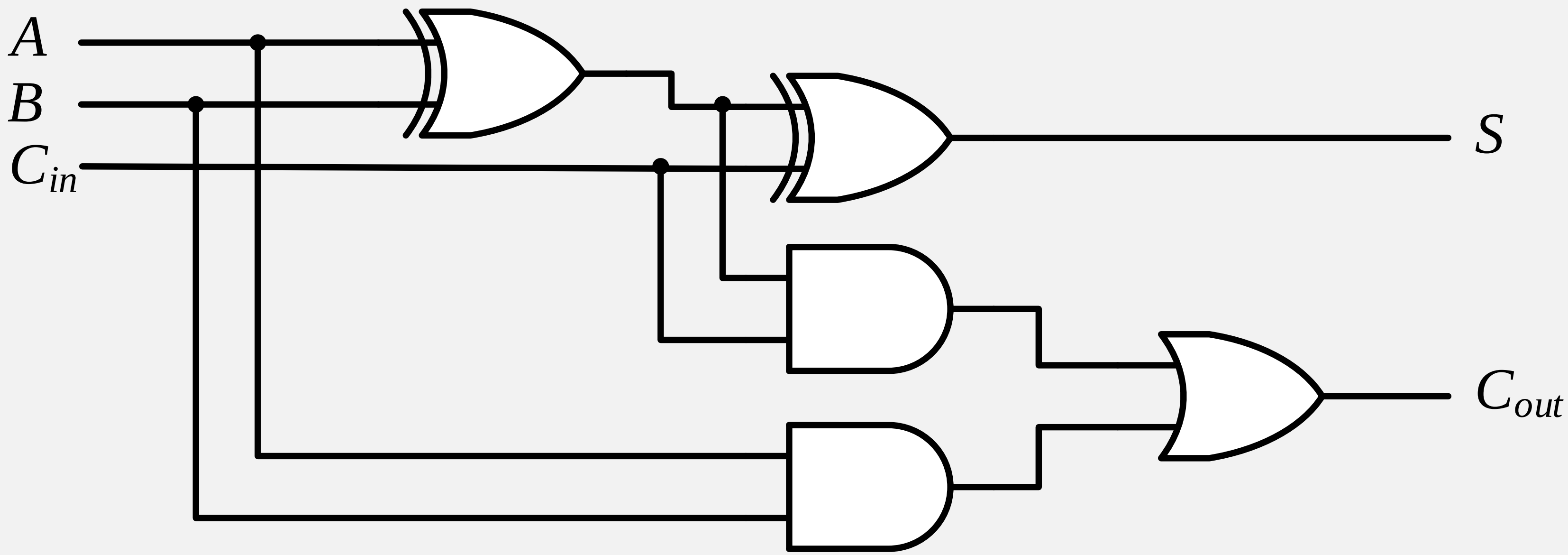
$$C_o = A\cdot B + A\cdot \overline{B}\cdot C_i + \overline{A}\cdot B\cdot C_i = A\cdot B + (A\cdot \overline{B} + \overline{A}\cdot B)\cdot C_i = A\cdot B + (A\oplus B)\cdot C_i$$

Gate Implementation of FA

- Logic equation

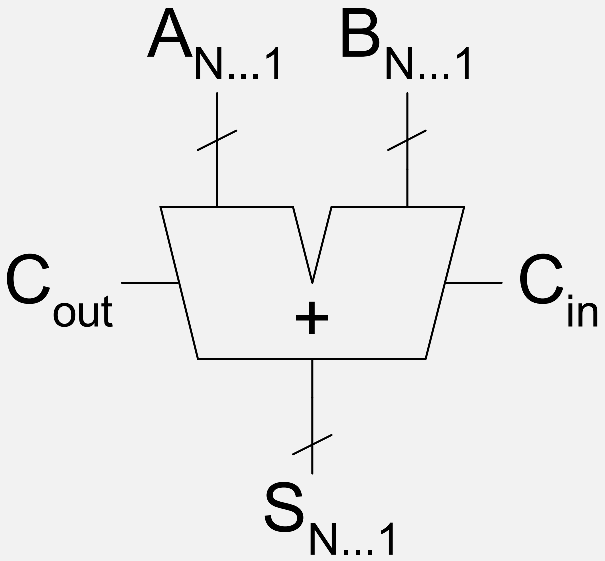
$$S = A \oplus B \oplus C_i$$

$$C_o = A \cdot B + (A \oplus B) \cdot C_i$$



Carry Propagate Adders

- Carry propagate adder
  - Each sum bit depends on all previous carries

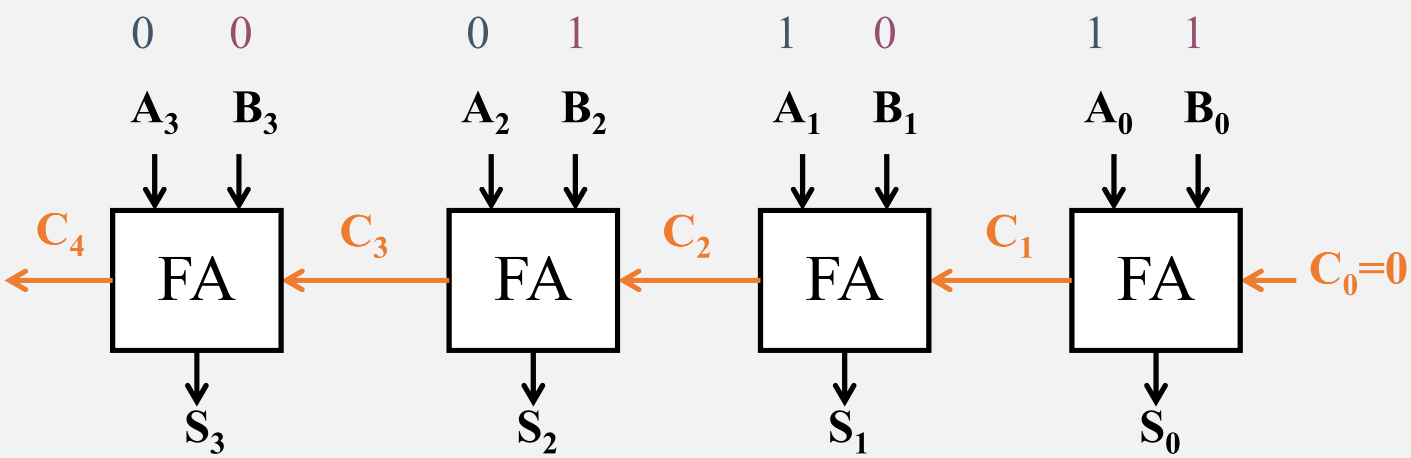


	C_{out} C_{in}	C_{out} C_{in}	
	00000	11111	carries
	1111	1111	$A_{4...1}$
	+0000	+0000	$B_{4...1}$
	1111	0000	$S_{4...1}$

Example: 4-bit Ripple-Carry Adder

A=0011

B=0101



T=0	0	0	0	0	0	0	0	0	S=0000
T=1	0	0	0	1	0	1	$\boxed{1}$	0	S=0110
T=2	0	0	0	1	$\boxed{1}$	0	1	0	S=0100
T=3	0	0	$\boxed{1}$	0	1	0	1	0	S=0000
T=4	0	1	1	0	1	0	1	0	S=1000

Binary Multiplication

上海科技大学
ShanghaiTech University

➢ Example:

$$
\begin{array}{r l l}
1100 & : 12_{10} & \text{multiplicand} \\
\underline{0101} & : 5_{10} & \text{multiplier} \\
1100 & & \\
0000 & & \text{partial} \\
1100 & & \text{products} \\
\underline{0000} & & \\
00111100 & : 60_{10} & \text{product}
\end{array}
$$

➢ 4 x 4-bit multiplication

❖ Produce 4 4-bit partial products

❖ Sum these to produce 8-bit product

Binary Multiplication: General Form

- Multiplicand: $Y = (y_{N-1}, y_{N-2}, \dots, y_1, y_0)$
- Multiplier: $X = (x_{M-1}, x_{M-2}, \dots, x_1, x_0)$
- Product: $$P = \left(\sum_{j=0}^{N-1} y_j 2^j\right)\left(\sum_{i=0}^{M-1} x_i 2^i\right) = \sum_{i=0}^{M-1}\sum_{j=0}^{N-1} x_i y_j 2^{i+j}$$

						y_5	y_4	y_3	y_2	y_1	y_0	multiplicand
						x_5	x_4	x_3	x_2	x_1	x_0	multiplier
						x_0y_5	x_0y_4	x_0y_3	x_0y_2	x_0y_1	x_0y_0	
					x_1y_5	x_1y_4	x_1y_3	x_1y_2	x_1y_1	x_1y_0		
				x_2y_5	x_2y_4	x_2y_3	x_2y_2	x_2y_1	x_2y_0			partial products
			x_3y_5	x_3y_4	x_3y_3	x_3y_2	x_3y_1	x_3y_0				
		x_4y_5	x_4y_4	x_4y_3	x_4y_2	x_4y_1	x_4y_0					
	x_5y_5	x_5y_4	x_5y_3	x_5y_2	x_5y_1	x_5y_0						
p_{11}	p_{10}	p_9	p_8	p_7	p_6	p_5	p_4	p_3	p_2	p_1	p_0	product

Binary Multiplication:An Example

- Multiplicand: 101010
- Multiplier: 1011

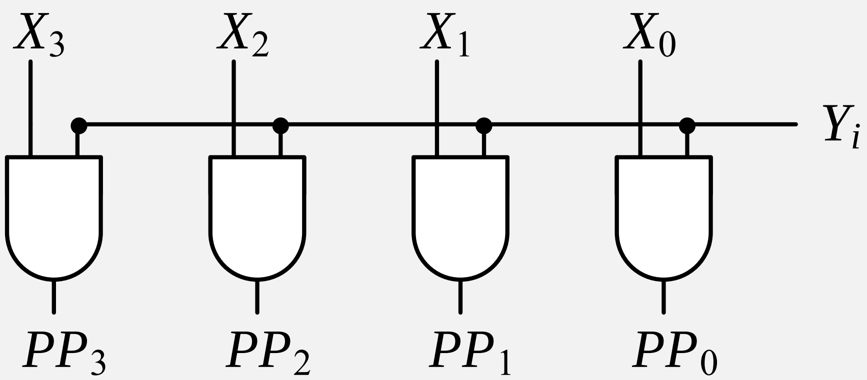
				1	0	1	0	1	0	Multiplicand
x						1	0	1	1	Multiplier
				1	0	1	0	1	0	Partial products
			1	0	1	0	1	0		
		0	0	0	0	0	0			
+	1	0	1	0	1	0				
	1	1	1	0	0	1	1	1	0	Result

Partial Product Generation

- Logical AND of multiplicand X with a bit Y_i

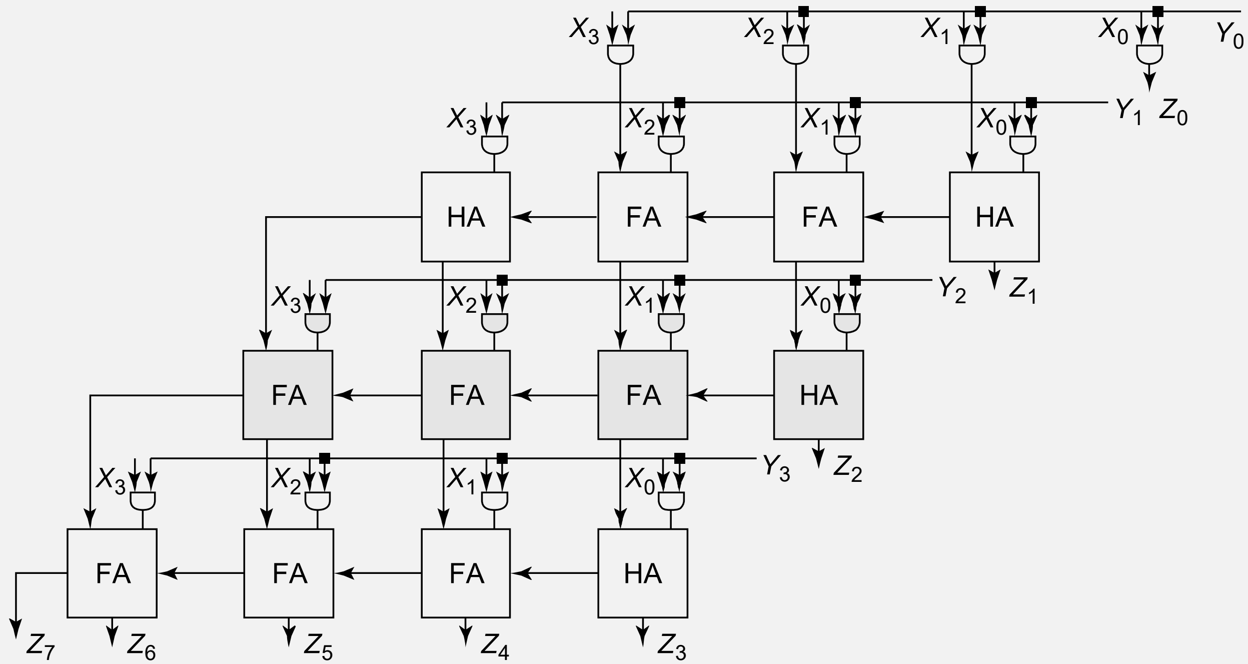
$$PP = X \times Y_i = \left(X_{M-1}X_{M-2}X_{M-3}\ldots X_1X_0\right)\left(Y_i\right)$$
$$= \left(X_{M-1}Y_i \quad X_{M-2}Y_i \quad X_{M-3}Y_i \quad \ldots X_1Y_i \quad X_0Y_i \quad\right)$$

- Example



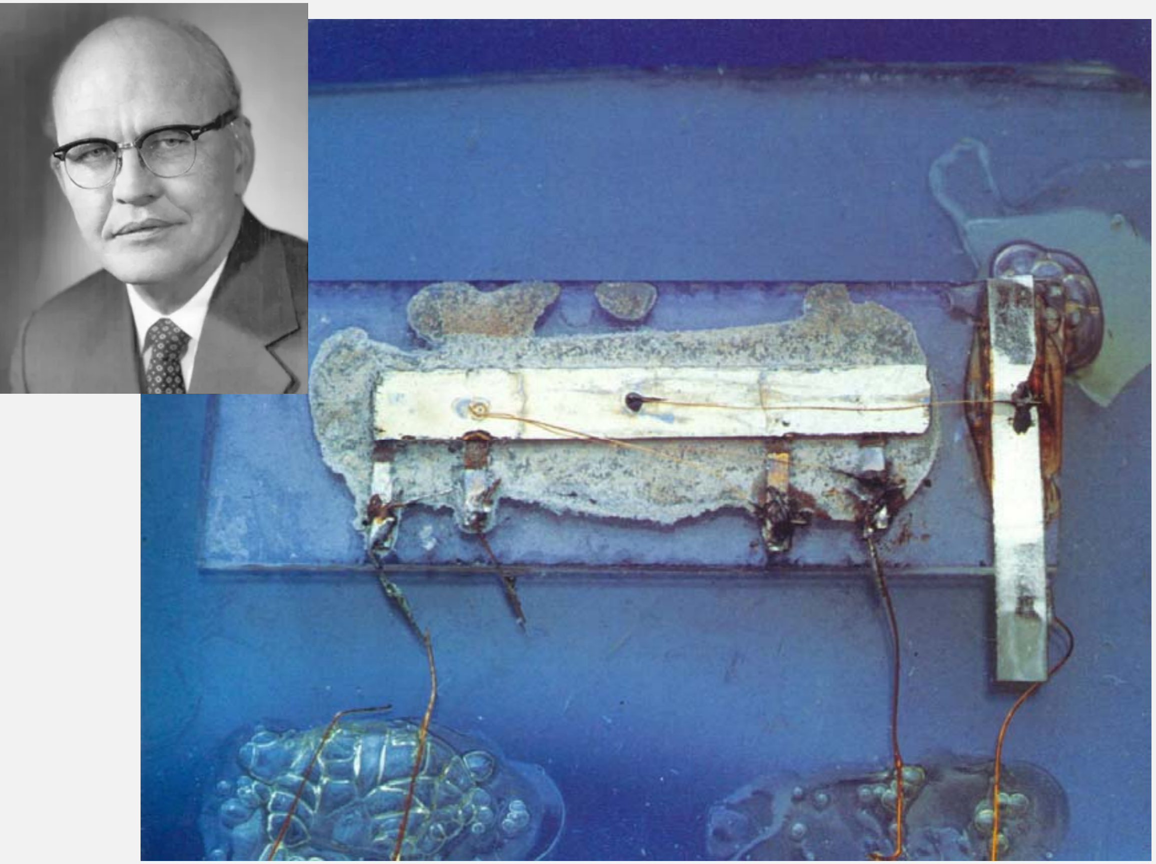
- Either a copy of X (Y_i=1) or a row of zeros (Y_i=0)

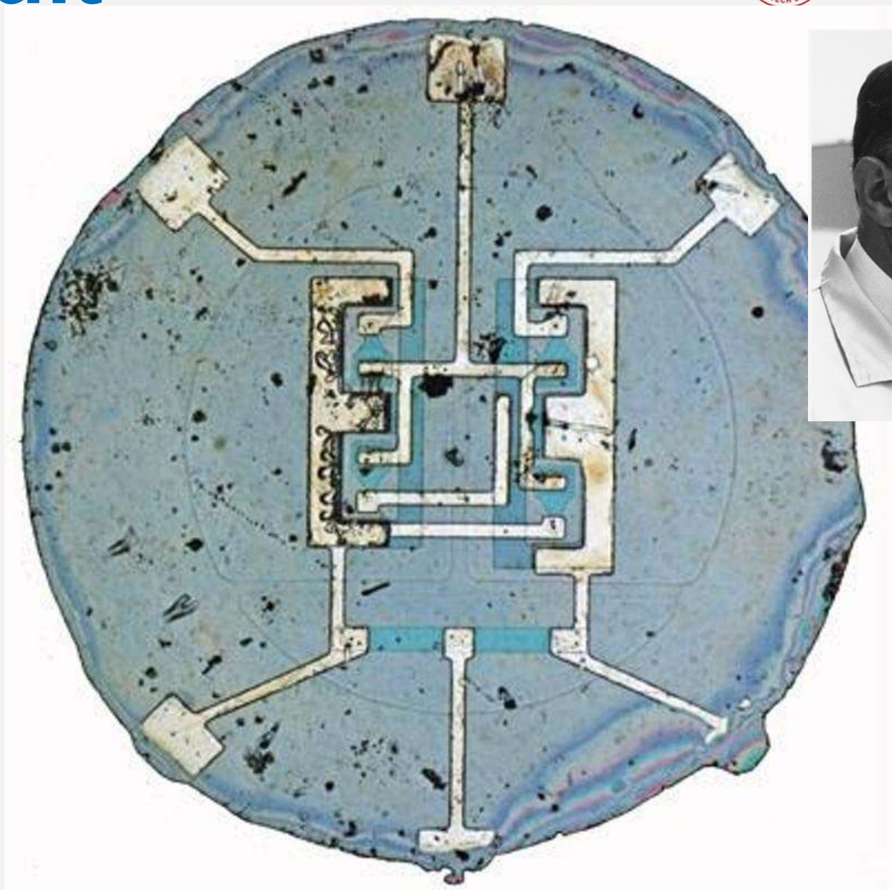
The Array Multiplier



- 4 partial products of 4 bit size each
- 4×4 two bit AND; 3 4-bit adders

The First Integrated Circuit





Jack Kilby, Texas Instruments, 1958

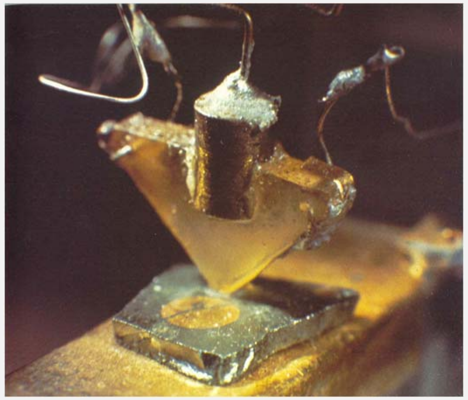
Nobel Prize for Physics in 2000

Robert Noyce, Fairchild

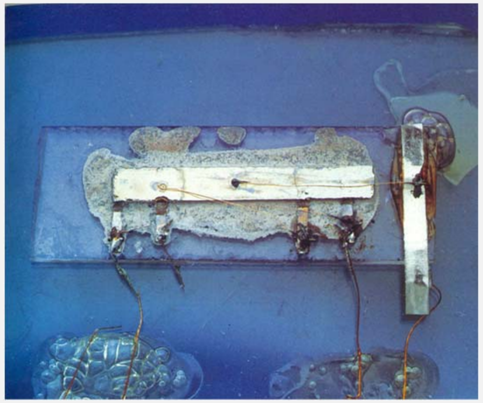
The Evolution

上海科技大学
ShanghaiTech University

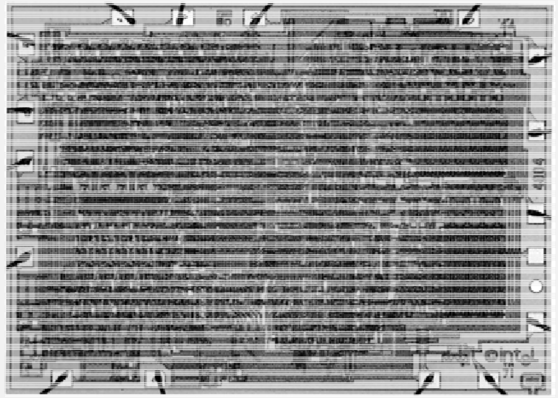
MOSFET scaling (process nodes)
•10 µm – 1971
•6 µm – 1974
•3 µm – 1977
•1.5 µm – 1981
•1 µm – 1984
•800 nm – 1987
•600 nm – 1990
•350 nm – 1993
•250 nm – 1996
•180 nm – 1999
•130 nm – 2001
•90 nm – 2003
•65 nm – 2005
•45 nm – 2007
•32 nm – 2009
•22 nm – 2012
•14 nm – 2014
•10 nm – 2016
•7 nm – 2018
•5 nm – 2020
•3 nm – 2022



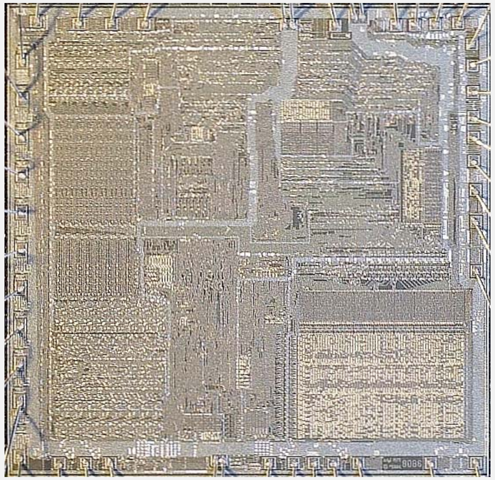
First transistor
Bell Labs, 1948



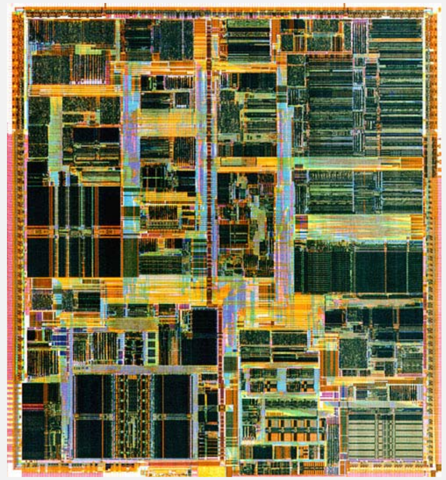
First IC, Jack Kilby
Texas Instruments, 1958



The Intel 4004,1971
~2250 transistors



The Intel 8086,1978
~29,000 transistors

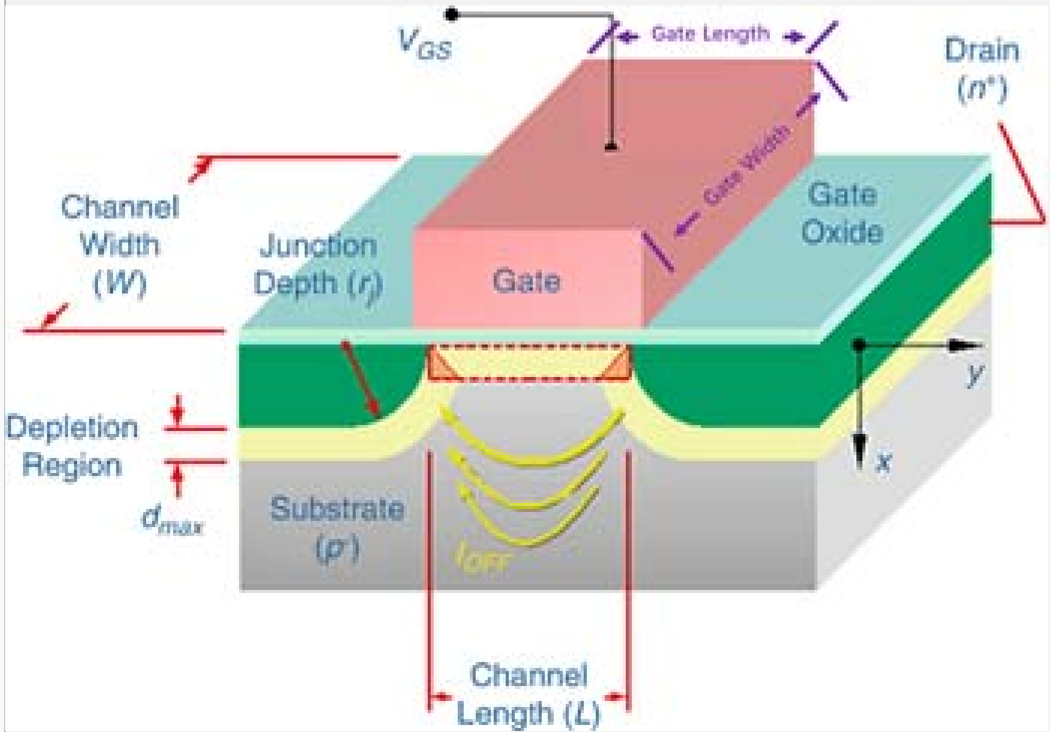


Intel Pentium IV, 2000
~42,000,000 transistors

Meaning of ? nm

MOSFET scaling (process nodes)
•10 µm – 1971
•6 µm – 1974
•3 µm – 1977
•1.5 µm – 1981
•1 µm – 1984
•800 nm – 1987
•600 nm – 1990
•350 nm – 1993
•250 nm – 1996
•180 nm – 1999
•130 nm – 2001
•90 nm – 2003
•65 nm – 2005
•45 nm – 2007
•32 nm – 2009
•22 nm – 2012
•14 nm – 2014
•10 nm – 2016
•7 nm – 2018
•5 nm – 2020
•3 nm – 2022

Previously



Now

技术的代号（营销）

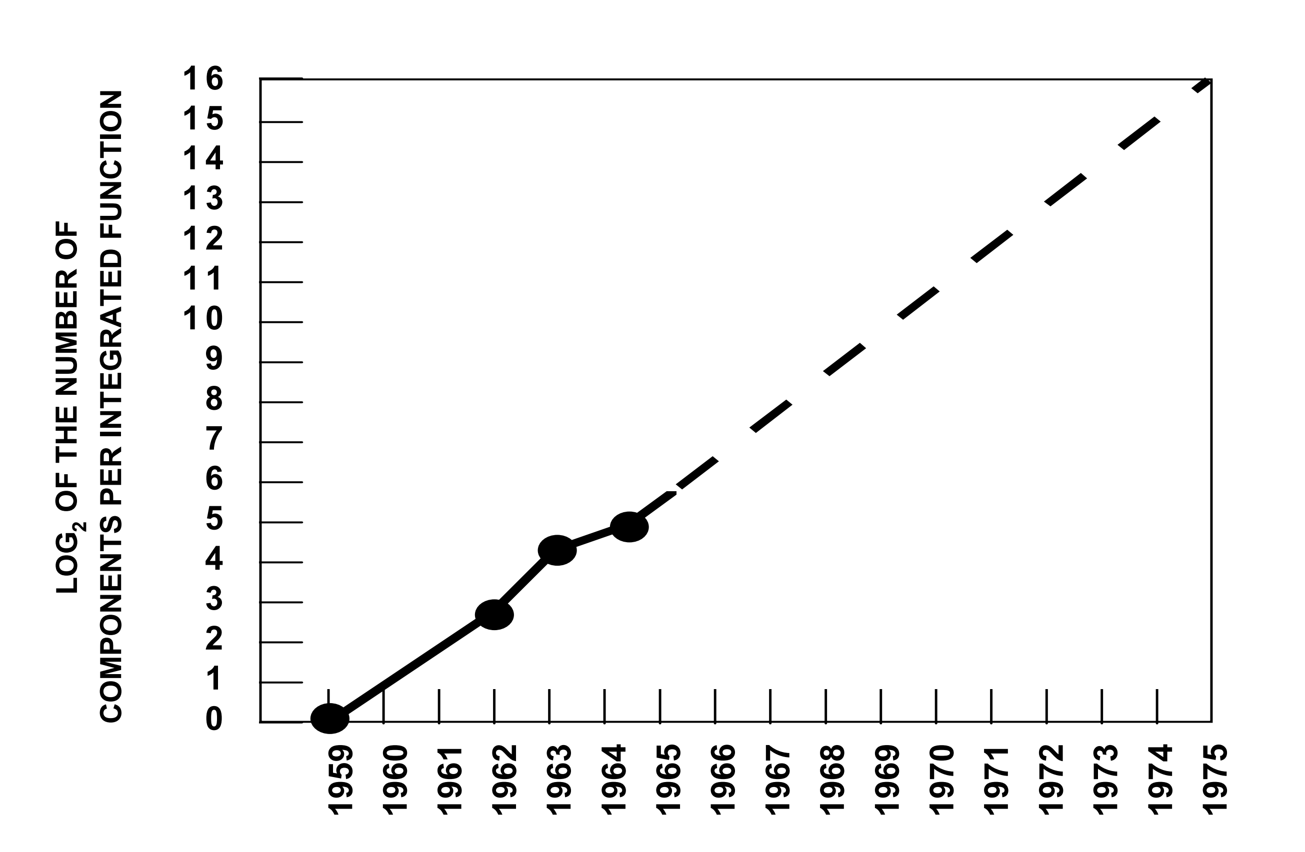
“It used to be the technology node, the node number, means something, some features on the wafer. Today, these numbers are just numbers. They’re like models in a car – it’s like BMW 5-series or Mazda 6. It doesn’t matter what the number is, it’s just a destination of the next technology, the name for it. So, let’s not confuse ourselves with the name of the node with what the technology actually offers.”

Philip Wong@HotChips31

Vice President of TSMC

The Moore's Law

In 1965, Gordon Moore noted that the number of transistors on a chip doubled every 18 to 24 months.



The Moore's Law

In 1965, Gordon Moore noted that the number of transistors on a chip doubled every 18 to 24 months.